



SCHOOL OF COMPUTER SCIENCE

Assisting First-Time Buyers in the Housing Market

A Comparative Analysis of Machine Learning Models for LSOA Value Forecasts

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Abstract

Housing is a critical social factor that has implications for both wellbeing and financial stability [8, 16]. First-time buyers represent a particularly vulnerable demographic, with their limited budget restricting them to buying the cheapest properties in desirable areas and forcing equity through renovation. This is a demanding strategy, and many first-time buyers lack both the resources and connections required to accomplish it. This paper constructs a machine learning pipeline to help first-time buyers identify undervalued areas with the highest growth potential over a three-year horizon, thus allowing the natural growth of the area to instead drive property equity.

The project achievements are outlined below:

- An extensive 30-year dataset (spanning 1995 to 2025) was compiled, consisting of financial data from all Lower Layer Super Output Areas (LSOAs) in England and Wales, along with spatial locality scores representing the impact of local facilities (schools, universities, business hubs, and transportation networks).
- Three regression models (Multiple Linear Regression, Geographically and Temporally Weighted Regression, and XGBoost) were evaluated alongside a state-of-the-art Spacetimeformer model adapted from traffic and weather forecasting.
- A direct classifier pipeline and regress-then-classify pipeline were constructed, based on the regression models, and then compared.
- An optimal pipeline for area value forecasting was identified, using an XGBoost direct classifier that captured two-thirds of all top 25% areas with a precision 1.6 times that of random guessing. This was complemented by an XGBoost regression model for granular area-level analysis.
- The adaptability of the XGBoost classifier at various risk levels was demonstrated through both a cautious configuration (79% precision) and an aggressive configuration (74% recall of top-quartile areas).

Dedication and Acknowledgements

Special thanks to my supervisor Dr James Cussens for keeping me on track and reviewing my draft. I would also like to thank Will Bowden for his helpful feedback on my drafts and for providing much needed moral support during our many long hours of independent study in MVB.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others including AI methods, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Sonny Cooper, Tuesday 5th May, 2026

AI Declaration

I declare that any and all AI usage within the project has been recorded and noted within Appendix A or within the main body of the text itself. This includes (but is not limited to) usage of text generation methods incl. LLMs, text summarisation methods, or image generation methods.

I understand that failing to divulge use of AI within my work counts as contract cheating and can result in a zero mark for the dissertation or even requiring me to withdraw from the University.

Sonny Cooper, Tuesday 5th May, 2026

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Ethics Statement

This project did not require ethical review, as determined by my supervisor, James Cussens

Supporting Technologies

- I used Python's `scikit-learn` library for the Multiple Linear Regression model, data preprocessing, and evaluation metrics.
- I used Python's `XGBoost` library for all XGBoost implementations.
- I adapted the open-source `spacetimeformer` library <https://github.com/QData/spacetimeformer> for the Spacetimeformer implementation.
- I used Python's `shap` library for feature importance analysis.
- I used Python's `pandas` and `NumPy` libraries for data manipulation and numerical operations.
- I used Python's `Matplotlib` and `Seaborn` libraries for creating visualisations.
- I used `pickle` to store datasets.
- I used R's `GWmodel` library to implement the Geographically and Temporally Weighted Regression (GTWR) model.
- I used Python's `argparse` library for parsing command-line arguments.
- I used the University of Bristol's BluePebble supercomputer to provide the computational resources required for the transformer and GTWR model training.

Notation and Acronyms

AIC	: Akaike Information Criterion
b	: Bandwidth
CBD	: Central Business District
CPI	: Consumer Price Inflation
DfE	: Department for Education
G_{yoy}	: Year-on-year percentage change in True Price
GNN	: Graph Neural Network
GTWR	: Geographically and Temporally Weighted Regression
GWR	: Geographically Weighted Regression
HESA	: Higher Education Statistics Agency
LAD	: Local Authority District
LPI	: Local Authority District Price Index
LSOA	: Lower Layer Super Output Area
MLR	: Multiple Linear Regression
MSE	: Mean Squared Error
NaPTAN	: National Public Transport Access Node
Norm	: Min-Max Scaling
ONS	: Office of National Statistics
PPD	: Price Paid Data
R^2	: Coefficient of Determination (R-squared)
RFE	: Recursive Feature Elimination
RMSE	: Root Mean Squared Error
SHAP	: SHapley Additive exPlanations
STLag	: Spatio-Temporal Lag
t	: Time (Quarterly)
T	: Time (Yearly)
TSF	: Time Series Forecasting
XGBoost	: Extreme Gradient Boosting

Chapter 1

Introduction

Housing is a critical social factor that has implications for both wellbeing and financial stability [8, 16]. Due to this, real estate pricing is a focal point in research. However, the existing literature typically approaches this problem from two distinct perspectives. First, the perspective of an estate agent, using models to perform more accurate house valuations based on the physical characteristics of the property alongside its location and neighbourhood considerations, therefore minimising the likelihood of undervaluing a property [29, 42]. Second, the perspective of the government, determining the current value within regions in order to allocate funding, here the problem takes the form of “if I add x facility to y area how will that impact house prices in the region” [30, 18, 35, 43].

With the wellbeing and financial concerns stated, it can be assumed that the most vulnerable group is first-time buyers, the focal group of this paper. Here, the limited budget leads this group to weigh the value of living in a desirable area and living in a desirable house. As is shown by common phrases given to this group: “buy the worst house on the best street”, “location, location, location” and “gentrifiers’ intuition”, forced equity is the primary method suggested. However, this is an active strategy that requires significant capital and time. Since many first-time buyers lack these resources, a model that evaluates price appreciation by area, rather than by specific property, offers greater utility. This approach allows buyers to identify undervalued regions with high potential for natural growth, while providing a compounding effect for those who do choose to pursue forced equity.

In order to shift the problem into the context of a buyer, this paper proposes a more long-sighted approach. Moving from the current methodologies of static price evaluation from a provided point-in-time towards a dynamic percentage increase evaluation over a three-year horizon, thereby allowing for a more strategic analysis suited to a buyer (the stakeholder with the greatest long-term impact of an exchange).

This study will use both financial data and spatial features from all Lower Layer Super Output Areas (LSOAs) [27] in England and Wales to compare the performance of four different machine learning models in predicting a percentage increase over a three-year horizon for both known and unknown regions. These four models are as follows: Multiple Linear Regression (MLR), Geographically and Temporally Weighted Regression (GTWR), XGBoost, and Spacetimeformer.

Finally, to shift this regression-based evaluation into context that a first-time buyer can understand, this paper will compare the performance of various classifier architectures (both staged and direct). Areas will be classified according to their predicted value compared to all other areas within England and Wales, with the top 25% of areas in a given year falling into class 1 and all other areas into class 0.

The high-level objective of this project is to create a predictive model that can assist first-time buyers in England and Wales in determining suitable areas that will soon exhibit the highest rates of natural property value growth in the country. More specifically, the concrete aims are:

1. Develop an extensive dataset between the years 1995 and 2025, consisting of all financial and spatial attributes of England and Wales that are required to accurately predict real estate trends.
2. Implement four regression based predictors, determined through survey literature of plausible models exhibiting either previous examples of high performance in similar tasks or state of the art methodologies from parallel fields.
3. Explore multiple classification pipelines that can identify the top 25% of LSOAs in England and Wales.
4. Evaluate the performance of both the regression and classification components of the pipeline along with the impact of each feature.

-
5. Show how the final proposed pipeline is suitable for assisting first-time buyers in real estate.

Chapter 2

Contextual Background

2.1 Factors of a Housing Market

The housing market is not a singular uniform market, but rather a set of sub-markets, each with its own individual supply and demand curves and equilibrium points [43, 38]. This is shown in Fig 2.1, where although there are spatial impacts causing similarities, overall, the sub-markets are individual in their trends. We can therefore split the factors into two primary groups. First, the local factors that play a significant role in the prediction of an individual area's price. Second, the macroeconomic factors that impact the whole country through new policies and laws that ultimately shift the incentive for buying over renting in all sub-markets.

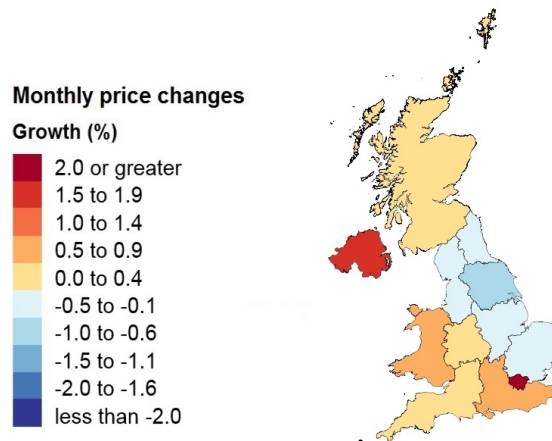


Figure 2.1: Monthly price changes by government office region 2025 [40]

2.1.1 Local Factors

The majority of housing prediction articles agree that there are three primary attributes that dictate housing prices: physical characteristics, location attributes, and neighbourhood considerations [35, 43, 44]. Expanding on this, recent studies now claim that spatial impacts, such as distances from facilities, along with the proximity to other up-and-coming areas are also key in determining more accurate predictions [35, 18].

Physical Characteristics

Physical characteristics focus on the individual building. Here, most characteristics show linear trends, with the number of rooms and dwelling area having a negative effect on price per square meter, whilst building quality has a positive effect. However, building age displays a non-linear trend with property ages up to 30 years having a detrimental effect, after which there is a positive effect [35].

Geographic culture, in addition to location attributes, has a significant impact on the perceived importance of physical characteristics for the buyer. For example, in China, south-facing buildings are one of the most critical factors due to the orientation of houses and apartments being closely associated with daylight exposure and the tradition of Fengshui [18]. Secondly, some articles have found that housing age has more influence on apartments in the inner-city, with urban areas prioritising private and public service amenities [18].

Location Attributes

Location attributes measure the ease of access to key facilities, with the Euclidean distance to the Central Business District (CBD), schools, universities, hospitals, parks, and main shopping centres having a significant impact on property prices in previous research [43, 35, 44]. Facilities such as hospitals tend to have a negative impact within 200 meters of the facility due to noise pollution and the increased traffic caused by ambulances [18, 35]. The quality of such facilities can also affect their perceived importance, with Li et al. [18] taking the ratings of local schools into account. Finally, it is important to note that some articles have argued that the significance of distance to a CBD is overemphasised [18].

In addition to this, access to transportation plays an enormous role, with Gray et al. [11] claiming that in the UK, for the period between 2010 and 2019, transport was the primary cause of gentrification within 33% of all instances. However, some papers have noted that the impact transportation and key facilities have in an area is partially dictated by its substitution effect, i.e. a bus route will have less of an impact on an area that already has a train line [43]. It is also important to understand that the various transportation services provide varying levels of impact, with trains having a greater positive impact than buses [18] and highway development having a significantly negative impact with its increased levels of noise pollution and traffic intensity [18].

Culture again affects the importance of these attributes as in the United States social differentiation is often related to racial and ethnic segmentation, while in China residential differentiation is dominated by occupational and income segmentation [18]. Consequently, attributes such as distance to the CBD will vary in impact based on the culture, with closer distances to the CBD in China likely to have a larger impact due to their segmentation being based primarily on occupation.

Neighbourhood Considerations

Neighbourhood and amenity considerations describe the quality of the area in which a house resides, although more subjective, aspects such as crime rates, pleasant scenery, and a quiet atmosphere are often attributes taken into account [44, 35, 18, 43]. Although area quality is harder to define, the attributes listed are indicators of the potential quality of life of an area. It stands to reason that many people highly value an area they deem safe, clean, and beautiful; this is reflected in the market prices of these areas.

Spatio-Temporal Lag

Although many papers have implemented some form of spatial correlation attribute that makes use of the distance to various key facilities [43, 35], recent articles claim that the use of a spatial lag attribute is key in improving a model's predictive capabilities [35, 18]. Spatial lag uses the prices at which nearby properties were sold as a metric to predict the potential price of the property they are aiming to sell. Equation 2.1 describes the formula for spatial lag as described by Soltani et al. [35]. Here, ST_i refers to the spatio-temporal lag for a property value i , $sd_{[i,j]}$ refers to the spatial distance between i and j , $td_{[i,j]}$ refers to the temporal distance between i and j , r_t refers to temporal search radius, r_s refers to the spatial search radius, and hp_j refers to property value of j (in price per square meter).

$$ST_i = \frac{\sum_{j|sd_{[i,j]} < r_s; td_{[i,j]} < r_t} \frac{hp_j}{sd_{[i,j]}}}{\sum_{j|sd_{[i,j]} < r_s; td_{[i,j]} < r_t} \frac{1}{sd_{[i,j]}}} \quad (2.1)$$

2.1.2 Macroeconomic Factors

Although studies agree that macroeconomic factors have a major impact on house price prediction, they are rarely implemented in the majority of predictive models due to the inherent difficulty in defining and measuring such metrics [44]. Unlike other metrics shown in section 2.1.1, macroeconomic factors can affect not only the demand side of the market but also the supply side.

Land use and construction cost can alter the development of new properties and, with it, the supply [38]. Land-use regulations are shown to raise housing and developed land prices in both a locality and its adjacent areas [32]. With this, we can assume that when governments alter land-use regulations to accelerate the development of properties, this will be reflected in the current property prices of an area. Construction costs can increase due to factors such as banks' increased interest rates due to inflation or the labour market trends increasing or decreasing the number of available labourers. With increased construction costs, supply can decrease; showcasing both the fragility of the market to macroeconomic factors and the difficulty that comes with monitoring such effects due to the multitude of potential sources of effect.

On the demand side, migration patterns, population growth, and income levels have a significant impact [38]. However, these effects are more easily monitored, with many using population as a key metric within their models [35, 2, 43]. Migration patterns are massively affected by macroeconomic factors like unemployment rates and geopolitics such as world conflicts forcing people to seek asylum. As the population of an area increases, so does the property values in the area. This relationship is a primary driver for the development of many predictive models [38, 35, 18], as understanding the trends of the housing market can allow the government to control housing issues through methods such as strategic placement of facilities, transportation, and new properties.

The study by Case et al. [2] is an older paper with a primary focus on the macroeconomic effects on housing prices. They found that sharp declines in marginal tax rates had a significant effect on whether an owner would choose to buy a property over renting. In times when it was more beneficial to buy demand rose and with it the price. When there were changes in the depreciation rules, the ITC, and passive loss rules, many owners instead favoured renting, decreasing the buying price of properties. This further solidifies the concept that macroeconomic impacts, such as government policies, have a substantial effect on house prices. Case et al. proves that a viable model can be created primarily through macroeconomic factors.

2.2 Area Classification in the UK

2.2.1 Regions

Since the unification of the Kingdom of England, regions have mainly been logical divides for administrative, statistical, and defensive purposes. Evolving from the 1066 system of ancient counties sharing sheriffs, the regional structure eventually led to the creation of Civil Defence Regions. Here, the UK was strategically divided into 12 regions for defensive purposes during World War II. A further shift came in 1945 in which the UK was split into Standard Regions; an 8-region split was used for economic planning and census consistency. Finally, the division of England into 9 regions, known as the Government Office Regions, came about in 1994, remaining the current default statistical regions.

Table 2.1: Official Government Office Region Codes

<i>Official Code</i>	<i>Region</i>	<i>Official Name</i>	<i>Country</i>	<i>#LADs</i>
E12000001		North East	England	12
E12000002		North West	England	39
E12000003	Yorkshire and The Humber		England	21
E12000004		East Midlands	England	35
E12000005		West Midlands	England	30
E12000006		East of England	England	45
E12000007		London	England	33
E12000008		South East	England	64
E12000009		South West	England	30
N92000002		Northern Ireland	Northern Ireland	11
S92000003		Scotland	Scotland	32
W92000004		Wales	Wales	22

With England split into 9 regions, codes for these regions, along with codes for Scotland, Wales, and Northern Ireland, were then assigned to allow for convenient statistical evaluations [28] (shown in

Table 2.1 and Fig 2.1). Because population and migration tracking utilise these codes alongside public spending allocation, it is a logical high-level abstraction and method of division to use in testing the robustness of models across unseen markets.

2.2.2 Local Authority Districts

Local Authority Districts (LADs) are a further subdivision of regions used for administrative and statistical purposes. This is the level responsible for local services such as planning, waste collection, and housing. Due to the local government’s control over housing, this is a typical level of abstraction used to represent a sub-market.

Overall, the UK consists of a total of 344 local authority districts [31], with London comprising 32 boroughs and the City of London. Wales, Scotland, and Northern Ireland possess consistent administrative structures through their respective Unitary Authorities, Council Areas, and District Councils. The distribution of these districts across each region is detailed in Table 2.1.

2.2.3 Lower Layer Super Output Areas

Lower Layer Super Output Areas (LSOAs) are a further level down from LADs in the geographic hierarchy for England and Wales, and are a single step up from the lowest level of geographical area for census statistics, being Output Areas. They represent geographic units composed of between 400 and 1,200 households and typically consist of four or five Output Areas. With a focus on consistent population size, LSOAs are redrawn every 10 years following a census to maintain population thresholds. Due to this, they are less stable than the higher levels of abstraction and are likely the lowest level of functional abstraction that can be used in a task such as the prediction of up-and-coming areas over a 3-year basis. As of 2021, there are 33,755 LSOAs in England and 1,917 in Wales [27].

Chapter 3

Technical Background

3.1 Regressive Model

3.1.1 Multiple Linear Regression

The following mathematical framework for Multiple Linear Regression is taken from the seminal work of Kutner, Nachtsheim, Neter and Li [17]. Multiple linear regression (MLR) is one of the simplest models we can implement when there are multiple predictor variables. Here, the model obtains the global average from all data in order to make predictions. If there is spatial stationarity, this model will be enough to produce accurate predictions for all locations. Equation 3.1 shows the first-order model for this, where β represents parameters, X are known constants, and ε are independent normal random variables ($\varepsilon \sim \mathcal{N}(0, \sigma^2)$).

$$y_i = \beta_0 + \sum_k \beta_k x_{ik} + \varepsilon_i \quad (3.1)$$

Although most of the data is quantitative, this method allows me to also measure qualitative predictive variables. A housing example of this is shown in Equation 3.2.

$$X_2 = \begin{cases} 1 & \text{if there is a garage} \\ 0 & \text{otherwise} \end{cases} \quad (3.2)$$

The matrix form of this, as shown in Equation 3.3, allows us to hide enormous computational complexities. Here $\mathbf{Y}$ is a vector of responses, $\boldsymbol{\beta}$ is a vector of parameters, $\mathbf{X}$ is a matrix of constants, and $\boldsymbol{\varepsilon}$ is a vector of independent normal random variables with expectation zero. With this, we can produce the least squares estimated regression coefficients (Equation 3.4), and therefore the Mean Squared Regression (MSR) and Mean Squared Error (MSE) as shown in Equation 3.8 and Equation 3.9 respectively.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \quad (3.3)$$

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (3.4)$$

$$SSR = \mathbf{b}^T \mathbf{X}^T \mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}^T \mathbf{J} \mathbf{Y} \quad (3.5)$$

$$SSE = \mathbf{Y}^T \mathbf{Y} - \mathbf{b}^T \mathbf{X}^T \mathbf{Y} \quad (3.6)$$

$$SST = \mathbf{Y}^T \mathbf{Y} - \left(\frac{1}{n}\right) \mathbf{Y}^T \mathbf{J} \mathbf{Y} \quad (3.7)$$

$$MSR = \frac{SSR}{p-1} \quad (3.8)$$

$$MSE = \frac{SSE}{n-p} \quad (3.9)$$

Where $\mathbf{J}$ is an $n \times n$ matrix of 1s, and $p-1$ is the number of predictor variables.

Although MSE is useful for showing the variance of the residuals and MSR is useful for showing the predictive power of each feature on average, the true strength of calculating these variables is their

applications in model evaluation. The F -test indicates that the relationship between features and house prices is truly statistically significant, as shown in Equation 3.10, while R^2 shows the percentage of variation in the housing market the model actually captures, as shown in Equation 3.11.

$$F = \frac{MSR}{MSE} \quad (3.10)$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \quad (3.11)$$

3.1.2 GWR

Geographically Weighted Regression (GWR) is a form of regression that encodes spatial dependencies into its predictions. Due to this, many papers label it a potential model in house price prediction [43, 18, 35], with the original literature specifically providing a housing price example [9].

The following mathematical framework for Geographically Weighted Regression is taken from the seminal work of Fotheringham, Brunsdon, and Charlton [9]. GWR is an alternative linear regression method for the local analysis of relationships in multi-variate datasets [9]. This method incorporates local spatial relationships into a regressive framework in an intuitive and explicit manner. Although similar to MLR, GWR uses $\mathbf{W}$ to capture the fact that not every observation is weighted equally. The issue with this basic method comes from calibration, where if ordinary least squares regression is used to produce the parameter estimates it fails to capture the geospatial variation in features. Similarly, grouping by hard boundaries, such as London Boroughs, can capture an area's features provided that the boroughs are the true spatial boundaries; a moving window removes this assumption but is still limited, being a discontinuous technique whose accuracy depends on the window size and suffers from edge effects.

GWR is a solution to this problem with fixed spatial kernels working by heavily weighting data points closer to the regression point, essentially smoothing the boundary found in the moving window method. However, this performs poorly when data is sparse [9], and so adaptive spatial kernels are instead implemented, adapting the kernel size according to the level of sparsity. Unlike global values, which are essentially spatial averages, GWR allows us to capture the geospatial variance within features. GWR reduces the spatial autocorrelation in the residuals and is less extreme than global models [9], proving itself a good contender in housing market models.

$$y_i = \beta_0 + \sum_k \beta_k(u_i, v_i)x_{ik} + \varepsilon_i \quad (3.12)$$

Equation 3.12 [9] is the equation for GWR; although similar to Equation 3.1 it makes use of variables u and v as coordinates of the i^{th} point in space. The issue that arises from this basic equation is that there are more unknowns than observed variables, meaning an unbiased estimate of local coefficients is not possible. However, calibration can still take place with a small amount of bias, provided the local sample is large enough. With the ordinary least squares method, as the subset gets larger, the chance of a coefficient drift increases, causing more bias [9]. Due to this, weighted least squares is the basis for GWR.

$$\begin{aligned} \hat{\beta}(u_i, v_i) &= \mathbf{C}\mathbf{y} \\ &= (\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{y} \end{aligned} \quad (3.13)$$

The weighted least squares used for calibration can be seen in Equation 3.13 [9], where $\mathbf{W}$ is an $n \times n$ diagonal matrix of the geographical weightings of each n observed data point for the regression point i . Finding the standard errors, we can square root the variance of this equation, as seen in Equation 3.14 [9].

$$SE(\hat{\beta}_i) = \sqrt{Var(\hat{\beta}_i)} \quad (3.14)$$

$$Var[\hat{\beta}(u_i, v_i)] = \mathbf{C}\mathbf{C}^T \sigma^2 \quad (3.15)$$

σ^2 , shown in Equation 3.15 [9], denotes the normalised residual sum of squares from local regression:

$$\hat{\sigma}^2 = \frac{\sum_i (y_i - \hat{y}_i)}{(n - 2\nu_1 + \nu_2)} \quad (3.16)$$

Where $\nu_1 = \text{tr}(\mathbf{S})$ and $\nu_2 = \text{tr}(\mathbf{S}^T \mathbf{S})$ respectively. Here, tr stands for trace and is the sum of its diagonal elements (top left to bottom right) and $\mathbf{S}$ represents a hat matrix, a mapping from the observed values to the fitted values. This hat matrix mapping is calculated using each row of $\mathbf{S}$, $\mathbf{r}_i$ given by:

$$\mathbf{r}_i = \mathbf{X}_i(\mathbf{X}^T \mathbf{W}(u_i, v_i) \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i) \quad (3.17)$$

With the weighted least squares function defined along with its standard error, which is required for calibration, we can now look into the spatial weighting function that defines how the GWR behaves. w_{ij} , an individual weighting within $\mathbf{W}$, can be implicit, such as when splitting by borough. This can be defined as a moving window or by the GWR method briefly hinted above. Here we use the formula for the nearest neighbour, as seen in Equation 3.18 [9] and Equation 3.19 [9]. Note the variable b , this denotes the bandwidth; a variable which does not implicitly relate to the model itself but a part of the calibration strategy for a given sample. As we tune the bandwidth we can alter the bias-variance trade-off. Zero bias does not generate an optimal estimator as a reduction in bias can lead to an increase in variance, so the use of cross validation scores is typically a means of selecting an optimal bias-variance combination for performance [9].

$$w_{ij} = \begin{cases} 1 & \text{if } j \text{ is one of the } N^{th} \text{ nearest neighbours of } i \\ 0 & \text{otherwise} \end{cases} \quad (3.18)$$

$$w_{ij} = \begin{cases} [1 - (d_{ij}/b)^2]^2 & \text{if } j \text{ is one of the } N^{th} \text{ nearest neighbours of } i \text{ and } d \\ 0 & \text{is the distance to the } N^{th} \text{ nearest neighbour} \\ 0 & \text{otherwise} \end{cases} \quad (3.19)$$

Finally, to calibrate the spatial weighting function, we can derive the bandwidth through minimising the Akaike Information Criterion (AIC). This is currently a method that holds promise in GWR [9]. Equation 3.20 [9] shows this, where n is the sample size, $\hat{\sigma}^2$ is the estimated standard deviation of the error term, and $\text{tr}(\mathbf{S})$ denotes the trace of the hat matrix (see Equation 3.17 [9]).

$$AIC_c = 2n \log_e(\hat{\sigma}^2) + n \log_e(2\pi) + n \left\{ \frac{n + \text{tr}(\mathbf{S})}{n - 2 - \text{tr}(\mathbf{S})} \right\} \quad (3.20)$$

We can automate the minimisation of AIC through an optimisation technique such as a Golden Section search. This involves evaluating the function with three different values, such as a , b , and c , where $a < b < c$, then again at a new value d , with the objective of choosing a value which lies between a and b or b and c , discarding a or c to form a new triplet. Value b aims to be chosen such that the ratio of distances is approximately 0.618 (the golden ratio) times the distance between a and c . This process continues until two successive values are almost the same. The one issue with this method is that it may fall into a local minimum rather than the global minimum. However, in GWR, AIC functions are reasonably well behaved in this regard [9].

3.1.3 GTWR

The following mathematical framework for Geographically and Temporally Weighted Regression is taken from the seminal work of Huang, Wu, and Barry [15]. Geographically and Temporally Weighted Regression (GTWR) aims to extend the traditional GWR model to incorporate both spatial and temporal non-stationarity in real estate data. Rather than constructing a weight matrix based on distances between points, then separately accommodating for the time variable by adjusting the sale prices to match observations to a common date, GTWR constructs the weight matrix based on distances determined from (x, y, t) coordinates. The adapted formulae are shown in Equation 3.21 and Equation 3.22.

$$y_i = \beta_0(u_i, v_i, t_i) + \sum_k \beta_k(u_i, v_i, t_i) x_{ik} + \varepsilon_i \quad (3.21)$$

$$\hat{\beta}(u_i, v_i, t_i) = (\mathbf{X}^T \mathbf{W}(u_i, v_i, t_i) \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}(u_i, v_i, t_i) \mathbf{y} \quad (3.22)$$

Where $\mathbf{W}(u_i, v_i, t_i) = \text{diag}(w_{i1}, w_{i2}, \dots, w_{in})$. Here $w_{ij}(1 \leq j \leq n)$ are space-time distance functions of (u, v, t) , thus, the GTWR relies on the appropriate specification of the space-time distance decay function w_{ij} .

The primary challenge of this model is determining from the three-dimensional space-time coordinate system what is deemed close and therefore what should have a greater influence. The issue stems from

the fact that location and time are usually measured in different units and therefore have different scale effects. Due to this, an ellipsoidal coordinate system is needed to measure the closeness between a regression point and its surrounding observed points. Equation 3.23 shows how this is split, with λ and μ being scale factors used to balance the different effects to ensure neither effect degrades the other.

$$d^{ST} = \lambda d^S + \mu d^T \quad (3.23)$$

With Equation 3.24 providing the squared distance through time and spatial parameters along with scaling factors, we can define the components of W 's diagonal matrix by using these distances and calculating their respective bandwidths, as shown in Equation 3.25.

$$(d_{ij}^{ST})^2 = \lambda[(u_i - u_j)^2 + (v_i - v_j)^2] + \mu(t_i - t_j)^2 \quad (3.24)$$

$$\begin{aligned} w_{ij} &= \exp \left\{ - \left(\frac{\lambda[(u_i - u_j)^2 + (v_i - v_j)^2] + \mu(t_i - t_j)^2}{b_{ST}^2} \right) \right\} \\ &= \exp \left\{ - \frac{(d_{ij}^S)^2}{b_S^2} \right\} \times \exp \left\{ - \frac{(d_{ij}^T)^2}{b_T^2} \right\} \\ &= w_{ij}^S \times w_{ij}^T \end{aligned} \quad (3.25)$$

Where $(d_{ij}^S)^2 = (u_i - u_j)^2 + (v_i - v_j)^2$, $(d_{ij}^T)^2 = (t_i - t_j)^2$, h_{ST} is a parameter of spatio-temporal bandwidth, and $b_S^2 = b_{ST}^2/\lambda$ and $b_T^2 = b_{ST}^2/\mu$ are parameters of the spatial and temporal bandwidths, respectively.

In the practical application of this model, an adaptive kernel is used. This modifies the bandwidth b_{ST} for each regression point i such that a constant number of nearest spatio-temporal neighbours is maintained, ensuring numerical stability in regions with sparse observations.

3.2 Gradient Boosted Tree Model

The following mathematical framework for the Gradient Boosted Tree model and the subsequent optimisations introduced by XGBoost presented in this section are derived from the foundational work of Chen and Guestrin [4]. Gradient boosted trees have been shown to give state-of-the-art results on many standard classification benchmarks. Consequently, XGBoost, a scalable machine learning system for tree boosting that has been used in previous housing papers [44], demonstrates promise in price prediction tasks, having provided a winning solution in store sales predictions [4].

Gradient boosted trees operate on the principle of additive training. Unlike bagging methods such as Random Forests, which reduce the variance by averaging independent trees in parallel, gradient boosted trees rely on gradient boosting, a tree ensemble method that reduces bias by fitting trees sequentially. Each successive tree f_t is designed to optimize parameters of an existing ensemble. Formally, $\hat{y}_i^{(t)}$ is the prediction of the i^{th} instance of the t^{th} iteration; here we aim to add f_t to minimise the objective [4]. Typically, this is done through a regularized learning objective function, such as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (3.26)$$

$$\text{where } \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

The above algorithm operates by adding a new function f_t , which best compensates for the errors of the previous iteration $\hat{y}_i^{(t-1)}$. By treating the model as a sum of functions, Equation 3.26 ensures that each step is a greedy optimisation aimed at minimising the specific loss l , while the regularisation term $\Omega(f_t)$ penalises overly complex structures to maintain generalisability.

By iteratively adding these weak learners, the ensemble gradually converges toward the global minimum of the loss function. This process can be viewed as gradient descent in the functional space, where each new tree represents a step in the direction that minimises the overall objective. XGBoost is one such gradient boosting tree method that introduces a regularised objective to balance the predictive power with model complexity to prevent the over-fitting typically associated with deep sequential ensembles.

3.2.1 XGBoost

XGBoost is a scalable tree boosting system that has a unified approach to handling all kinds of sparsity patterns (see Algorithm 3.3). It is a highly adaptive algorithm with methods such as out-of-core allowing it to process a lot more data than other competing models [4].

Input : I , instance set of node
Input : d , feature dimension
 $gain \leftarrow 0$
 $G \leftarrow \sum_{i \in I} g_i$, $H \leftarrow \sum_{i \in I} h_i$
for $k = 1$ **to** d **do**
 $G_L \leftarrow 0$, $H_L \leftarrow 0$
 for j **in** *sorted* (I , by x_{jk}) **do**
 $G_L \leftarrow G_L + g_j$, $H_L \leftarrow H_L + h_j$
 $G_R \leftarrow G - G_L$, $H_R \leftarrow H - H_L$
 $score \leftarrow$
 $\max(score, \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda})$
 end
end
Output : Split with max score
Algorithm 3.1: Exact Greedy Algorithm for Split Finding [4]

for $k = 1$ **to** m **do**
 Propose $S_k = \{s_{k1}, s_{k2}, \dots, s_{kl}\}$ by percentiles on feature k .
 Proposal can be done per tree (global), or per split (local).
end
for $k = 1$ **to** m **do**
 $G_{kv} \leftarrow \sum_{j \in \{j | s_{k,v} \geq x_{jk} > s_{k,v-1}\}} g_j$
 $H_{kv} \leftarrow \sum_{j \in \{j | s_{k,v} \geq x_{jk} > s_{k,v-1}\}} h_j$
end
 Follow same step as in previous section to find max score only among proposed splits.
Algorithm 3.2: Approximate Algorithm for Split Finding [4]

Input : I , instance set of current node
Input : $I_k = \{i \in I | x_{ik} \neq \text{missing}\}$
Input : d , feature dimension
Also applies to the approximate setting, only collect statistics of non-missing entries into buckets
 $gain \leftarrow 0$
 $G \leftarrow \sum_{i \in I} g_i$, $H \leftarrow \sum_{i \in I} h_i$
for $k = 1$ **to** m **do**
 // enumerate missing value goto right
 $G_L \leftarrow 0$, $H_L \leftarrow 0$
 for j **in** *sorted* (I_k , ascent order by x_{jk}) **do**
 $G_L \leftarrow G_L + g_j$, $H_L \leftarrow H_L + h_j$
 $G_R \leftarrow G - G_L$, $H_R \leftarrow H - H_L$
 $score \leftarrow \max(score, \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda})$
 end
 // enumerate missing value goto left
 $G_R \leftarrow 0$, $H_R \leftarrow 0$
 for j **in** *sorted* (I_k , descent order by x_{jk}) **do**
 $G_R \leftarrow G_R + g_j$, $H_R \leftarrow H_R + h_j$
 $G_L \leftarrow G - G_R$, $H_L \leftarrow H - H_R$
 $score \leftarrow \max(score, \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda})$
 end
end
Output: Split and default directions with max gain

Algorithm 3.3: Sparsity-aware Split Finding [4]

The key benefit of XGBoost is its scalability. Unlike traditional tree boosting, XGBoost uses second-order Taylor expansion in its loss function, shown in Equation 3.27. This algorithm simplifies the objective for parallelisation, causing the objective function to become independent from the loss used [4]. Benchmarks have shown XGBoost to perform up to ten times faster than alternative popular solutions while also scaling to billions of examples in distributed or memory-limited settings [4].

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^n [g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i)] + \Omega(f_t) \quad (3.27)$$

Here $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$ and $h_i = \partial_{\hat{y}^{(t-1)}}^2 l(y_i, \hat{y}^{(t-1)})$ are the first and second order gradient statistics on the loss function. It is important to note that since the loss is calculated point-wise for each instance, i , g_i and h_i are scalars rather than matrices, even in high-dimensional spaces. These are computed once at the beginning of each iteration t . By pre-calculating these values, the objective function in Equation 3.27 becomes a simple quadratic form of $f_t(\mathbf{x}_i)$. This allows the algorithm to find the optimal split through summing these scalar statistics (as shown in Algorithm 3.1), which significantly enhances computational efficiency and facilitates parallelisation. Solving this second-order equation for optimal leaf weights produces the following:

$$\tilde{\mathcal{L}}^{(t)} = \sum_{j=1}^T [(\sum_{i \in I_j} g_i) w_j + \frac{1}{2} (\sum_{i \in I_j} h_i + \lambda) w_j^2] + \gamma T \quad (3.28)$$

The fixed structure of a tree is defined by a mapping function $q : \mathbb{R}^d \rightarrow \{1, 2, \dots, T\}$, which assigns each data point to its corresponding leaf index based on its feature values. By defining the instance sets of leaf j as $I_j = \{i | q(\mathbf{x}_i) = j\}$, we can compute the optimal weight w_j^* of leaf j by:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (3.29)$$

Therefore, we can calculate the corresponding optimal value by:

$$\tilde{\mathcal{L}}^{(t)}(q) = -\frac{1}{2} \sum_{j=1}^T \frac{(\sum_{i \in I_j} g_i)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \quad (3.30)$$

Equation 3.30 ultimately gives us a scoring function for the tree function. By using this scoring function, rather than enumerating through all possible tree structures, which is normally impossible, we can use a greedy algorithm, such as Algorithm 3.1, to start from a single leaf and iteratively add branches to the tree. Equation 3.31 reflects that and is the typical practice for evaluating split candidates [4]. Note, γ essentially turns this into a pre-pruning method, acting as a threshold in deciding whether a split should be made.

$$\mathcal{L}_{split} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma \quad (3.31)$$

XGBoost uses two methods to further prevent over-fitting. Firstly, it uses shrinkage; this scales newly added weights by a factor of η after each step of tree boosting. This reduces the influence of each individual tree and leaves space for future trees to improve the model [4]. Secondly, it uses column (feature) subsampling; this is a technique used in RandomForests and is shown to prevent over-fitting even more than traditional row sub-sampling while also speeding up computations [4].

One property XGBoost has over many other gradient boosted tree models is its compatibility with multiple split finding algorithms. Both the more precise exact greedy algorithm, Algorithm 3.1, and the more distributed, memory-limited fallbacks of the approximate global algorithm and approximate local algorithm, Algorithm 3.2, can be used.

Overall, this computation speed, ability to parallelise and use of out-of-core computation makes XGBoost a good algorithm for large datasets such as the one used in the paper. Its ability to handle sparsity (sparse meaning the absence of data rather than geographical sparseness) further showcases its potential with it still contributing to the robustness of the system as the dataset passed does not need to be cleaned to the same extent as other methods.

3.3 Transformer Model

Transformer models are currently a popular area of deep learning, introduced by a paper in 2017 [41]. Transformers are a form of model that handle temporal forms of data, making them popular in tasks such as LLMs and temporal reasoning. Unlike other temporal models, such as RNNs, transformers rely on attention mechanisms to determine the relevance between all input tokens within the temporal sequence (shown in Equation 3.32) and positional embeddings (shown in Equation 3.33) to infer order rather than traditional time-stamps [41].

$$Attention(Q, K, V) = softmax \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3.32)$$

$$\begin{aligned} PE_{pos, 2i} &= \sin(pos/10000^{2i/d_{model}}) \\ PE_{pos, 2i+1} &= \cos(pos/10000^{2i/d_{model}}) \end{aligned} \quad (3.33)$$

In this paper, I will evaluate a particular cutting edge transformer model from 2023 that is tailored towards long-range spatio-temporal forecasting. This model has been shown to perform well in similar spatio-temporal tasks such as weather and traffic forecasting [12].

3.3.1 Spacetimeformer

The following details for the Spacetimeformer model presented in this section are derived from the foundational work of Grigsby et al. [12]. Spacetimeformer is a general-purpose multivariate forecaster with both the long-term prediction ability of a time series model and the dynamic spatial modelling of a Graph Neural Network (GNN) [12]. This is achieved by addressing the limitations of current state-of-the-art TSF (time series forecasting) models and the spatio-temporal structures found in GNNs. Firstly, current TSFs work by substituting the classic seq2seq architectures (See Appendix B) for neural-attention-based mechanisms. This leads to the model representing the value of multiple variables per time-step as a single input token, ignoring distinct spatial relationships. Secondly, spatio-temporal methods aiming to capture relationships between multiple variables achieve their goals through alternating temporal sequence processing and spatial message passing based on GNNs. These GNNs come as hardcoded graphs, making it difficult to define relationships that change over time.

Long-range transformers implement various strategies to lengthen token sequences. This is necessary as self-attention mechanisms determine the strength of every connection between the i^{th} token of a query sequence and the j^{th} token in the key sequence, leading to a massive adjacency matrix. Here, the size of the adjacency matrix can be defined as $A \in \mathbb{R}^{L_q \times L_k}$, where L_q is the length of the query sequence and L_k is the length of the key sequence. Time series forecasting (TSF) is a method based on the seq2seq framework, this maps a context window of the c most recent time-steps to a target window of predictions for a time horizon of h steps into the future [12]. Here, the time (x) and variable (y) components sum to create an input sequence of $c+h$ tokens, written in matrix form as $Z \in \mathbb{R}^{(c+h) \times d}$. After this, an encoder processes the sub-sequence $Z[0...c]$ while the decoder observes the target sequence $Z[c+1, ..., c+h]$. The outputs of this decoder are used as predictions, minimising the error relative to the ground truth values (via regression in the case of the seq2seq method). For the Spacetimeformer model, a transformer is instead used to minimise the error, using one token per time-step, so that the embedding of the token at time t represents N distinct variables at that moment in time. This setup forces variables within each token to receive the same amount of information from other time-steps [12].

Multivariable TSF involves two dimensions of complexity: the duration of the forecast sequences L , and the number of variables N considered at each time-step. With GNNs (a category of techniques that aim to explicitly model variable relationships as interactions along a network of nodes) for each time-step t , the graph $\mathcal{G}_t$ consisting of N variable nodes connected by weighted edges is constructed. Using the context sequence of the graphs $(\mathcal{G}_{T-c}, ..., \mathcal{G}_T)$, the target sequence $(\mathcal{G}_{T+1}, ..., \mathcal{G}_{T+h})$ can be predicted [12]. For a more dynamic approach, within spatio-temporal transformer models the matrix form $Z \in \mathbb{R}^{(c+h) \times d}$ is modified by flattening each multivariate y_t vector into N scalars with a copy of its timestamp x_t , becoming $Z' \in \mathbb{R}^{N(c+h) \times d}$. This produces a fully connected, learnable adjacency matrix $A \in \mathbb{R}^{N(c+h) \times N(c+h)}$, representing the spatio-temporal graph with a direct path between every time-step [12]. This means that learning spatial relationships no longer relies on predefined variable graphs and can instead change dynamically according to the time and variable values of the input data.

$$Z'_{i,t} = \underbrace{\text{Linear}(\text{Value}) + \text{Time2Vec}(\text{Time})}_{\text{value+time embedding}} + \text{Embed}(\text{Position}) + \text{Embed}(\text{Spatial}) \quad (3.34)$$

The embeddings of the Spacetimeformer are what drives the transformers ability to think in both time and space. This embedding is the concatenation of four components, the “value+time embedding”, the position embedding, and the spatial embedding. Firstly, the model captures the “value+time embedding” by creating an input sequence consisting of time series variables y and time information x . Using Time2Vec layers, seasonal characteristics are learned, mapping x_t to sinusoidal patterns of learned offsets and wavelengths to represent periodic relationships that extend beyond the limited length of the context sequence. The concatenated variable values and time embeddings are projected to the input dimension of the transformer model with a feed-forward layer [12]. Next, position embeddings are added to the tokens through initialising d -dimensional embedding vectors for each time-step up to the maximum sequence length. Finally, the spatial tokens are created by flattening the N variables into separate tokens. With each token, a copy of the time information x for its “value+time embedding” is given and the position embedding index corresponding to its original time-step order is assigned. The variables at each time-step are differentiated with additional embeddings, initialising N d -dimensional embedding vectors for each variable index [12]. The complete embedding logic is shown in Fig 3.1 and Equation 3.34.

One major limitation of this model is the size of the matrix $N(L) \times N(L)$. This is a quadratic complexity bottleneck of $O((NL)^2)$, Grigsby et al. [12] addresses this by supporting use of “Performer” (linear attention) or “Informer” (ProbSparse) mechanisms to ensure the model remains computationally

feasible as $N \times L$ becomes large.

Input Data				Temporal Sequence			
$y_{context}$	10	10	0	Value	10	10	0
	1	1	-9		1	1	-9
	-9	-8	-5		-9	-8	-5
$x_{context}$	T-c	T-1	T	Time	T-c	T-1	T
				Position	0	1	2

Spatiotemporal Sequence									
Value	10	10	0	1	1	-9	-9	-8	-5
Time	T-c	T-1	T	T-c	T-1	T	T-c	T-1	T
Spatial	0	0	0	1	1	1	2	2	2
Position	0	1	2	0	1	2	0	1	2

Figure 3.1: Example of Spacetimeformer embeddings [12]. (Top Left) An example of a context sequence. (Top Right) The standard “Temporal” embedding input sequence where each column will become a token. (Bottom) Flattened spatio-temporal input sequence whose columns will become a token through the process described in Equation 3.34

Chapter 4

Methodology and System Design

4.1 Outline

This section presents the development of the core components that make up the initial regression model used to predict the percentage increase in median house prices within LSOAs over a given three-year period. The sections of this chapter have been ordered to first describe the data wrangling and dataset construction, then the implementation of the models that use this dataset.

4.1.1 Methodology Overview

This paper will focus on data between 1995 and 2025 for England and Wales. The model will be allowed to use the previous three years to predict the percentage increase for three years in the future. The data will be split into three sections: predictions for 2001 to 2018 for training data, 2019 to 2021 for validation, and 2022 to 2025 for testing (a 70 : 15 : 15 split). Note that standard k -fold cross-validation could not be used due to the potential data leakage and look-ahead bias inherent in the strongly auto-correlated features in house pricing. Alongside this, two regions (the North West and London) have been removed from all training and validation, so an evaluation of performance on unknown regions may be taken. To maintain the integrity of the unknown regions the blocked temporal split was chosen over time-series forward chaining. The resulting dataset was then passed into the various models. Fig 4.1 describes the methodology and system design.

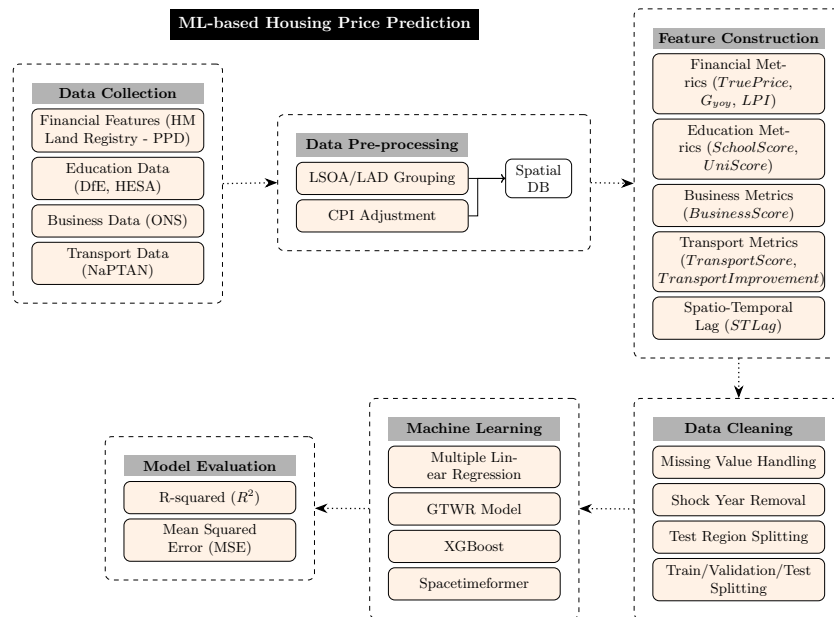


Figure 4.1: The flowchart of the methodology and system design process

4.1.2 Persona

In order to create a real-world robust system, I have decided to define a persona for the first-time buyer. With this, I can filter out properties and aspects that are likely not important to the first-time buyer to develop a more robust and precise model. The model assumptions are as follows:

- Will not buy brand new properties
- Will buy a house rather than a flat
- Aims to buy into up-and-coming areas

The first assumption that first-time buyers will not buy brand new stems from the fact that new builds are more costly, with the average price in November 2024 of a new build being £399,858, compared to the price of existing resold properties, which was £261,645 [40]. First-time buyers in January 2025 spent £103,691 less than former owner occupiers, at an average price of £227,026 [40]. This is substantially less than the average price of a new build, and implies that new builds are more targeted towards former owners who are already present on the property ladder.

Secondly, the assumption that the buyer will prioritise a type of house over a flat comes from the fact that a buyer who is interested in models that predict optimal areas likely wants to get a property with the highest potential value for the lowest cost. It's shown in the 2025 UK price index summary [40] that whilst the yearly increase in flats sat around 2.3%, detached, semi-detached, and terraced properties all sat between 4.7% – 6.0%, which is substantially higher, and so, if a buyer is purchasing a property with a focus on value appreciation they should choose a house over a flat.

4.2 Dataset Construction

The dataset consists of quarterly financial features, yearly spatial features, and a quarterly spatio-temporal lag feature, inspired by the spatio-temporal lag feature discussed by Soltani et al. [35] (see Section 2.1.1). In order to remove outliers, predictions for shock years such as 2008 – 2009 for the financial crisis and 2021 – 2022 for Covid-19 have been removed.

To create a spatial decay gravity model, used below for both the Spatial Features (Section 4.2.2) and spatio-temporal lag (Section 4.2.3), a KDTree was constructed (see Appendix B for details) due to its low time complexity in construction and retrieval. This tree takes the centroids of all LSOAs or LADs and constructs a tree in $O(n \log n)$ average time. Then, using a target LSOA/LAD and a defined radius, it returns a list of all neighbouring areas along with their distances from the target with a time complexity of $O(\sqrt{n} + m)$ where m is the number of points found.

4.2.1 Financial Features

Financial features are the basis of any housing market predictor with earlier articles, such as Case and Shiller [2] in 1990, relying fully on economic factors. Due to this, financial features will be the base metric on which the database is built. This is recorded on both the lowest level of abstraction for this model (LSOA) and the smallest time increments (quarterly) to ensure that transactional seasonality strength and price volatility can be captured. This section definition of time (t) will be at a quarterly level.

Financial features were extracted from the HM Land Registry - Price Paid Data (PPD) [14]. This dataset is updated every month and consists of every residential property sale in England and Wales from January 1995 to the current date. More specifically, it holds the following information about each sale:

- Transaction unique identifier
- Sale price
- Date of transfer
- Postcode
- Property type (detached, semi-detached, terraced, flat/maisonette, other)
- Old/new build
- Duration (freehold or leasehold)

- Primary addressable object name (PAON)
- Secondary addressable object name (SAON)
- Street
- Locality
- Town/City
- District
- County
- PPD category type (standard residential sale or additional price paid entry, e.g. repossessions, buy-to-lets)
- Record status (added, changed, deleted)

To align the dataset with the first-time buyer persona described in Section 4.1.2, new properties, flats, maisonettes, and other properties were all removed. Two new columns were then created, translating “date of transfer” into year and quarter. Finally, irrelevant columns were dropped, leaving the following: “Sale price”, “Quarter”, “Year”, “Postcode”.

Next, in order to translate the data into LSOA level information, the Office of National Statistics (ONS) LSOA lookup table was joined via the “Postcode” [23]. From this, the region, the LAD, and the individual LSOA values for 2001, 2011 and 2021 were joined (unifying the three different boundaries using the ONS LSOA lookup table to ensure that older spatial data can be translated into the newer boundaries). Finally, the dataset was grouped by quarter, year and LSOA. This translated individual residential sales into the median sales price of the LSOA and the volume of sales in the LSOA.

$$TruePrice_{LSOA,t} = Price_{LSOA,t} \cdot \frac{100}{CPI_{t,2015 \text{ adjusted}}} \quad (4.1)$$

Finally, in order to add context to the data and ensure it predicts house movement rather than the inflation of the pound, prices were adjusted using the ONS’s consumer price inflation (CPI) time series [25], with 2015 selected as the base year, as shown in Equation 4.1. Features describing the percentage change per year (Equation 4.2) and the LAD price index (Equation 4.3, where $TruePrice_{LAD,t}$ represents the mean $TruePrice_{LSOA,t}$ of all LSOAs within the corresponding LAD at time t) were calculated for each quarterly data-point to provide more relevant percentage level context.

$$G_{yoy,LSOA,t} = \frac{(TruePrice_{LSOA,t} - TruePrice_{LSOA,(t-4)})}{TruePrice_{LSOA,(t-4)}} \quad (4.2)$$

$$LPI_{LSOA,t} = \frac{TruePrice_{LSOA,t}}{TruePrice_{LAD,t}} \quad (4.3)$$

Following the extraction of these features, the “ground truth” targets, defined as the percentage return over a three-year horizon, were calculated as follows:

$$Return_{LSOA,3yr_t} = \frac{TruePrice_{LSOA,(t+12)} - TruePrice_{LSOA,t}}{TruePrice_{LSOA,t}} \quad (4.4)$$

4.2.2 Spatial Features

Unlike financial features, spatial features are a slower moving metric. Here, the time in this section will be defined as the T (uppercase) to show that it is at a yearly level.

Within the spatial features, there are no LSOA level scores using population size (university and business scores use population and run on a LAD level). This is because LSOAs have a consistent population size of 400 to 1200 households, adjusting their boundaries every ten years to ensure this remains consistent.

Certain traditional spatial metrics, more specifically hospitals and shopping centres, have been excluded from the final dataset. Hospitals have been omitted due to their focus on clinical activity over estate lifecycles, meaning, like university population metrics they are particularly hard to wrangle, often being formatted as a singular annual pdf. Due to both the challenge of wrangling this data and the fact

that hospital impact is found to have a non-linear relationship suggests it may be a less suitable LSOA level metric, with noise pollution and traffic congestion being detrimental to house prices within a very small distance of 200m [35]. It was therefore decided a more sophisticated university score was the more beneficial focus.

Similarly, shopping centres were excluded due to their status as private commercial assets. Unlike public infrastructure, there is no requirement for private developers to register precise start and end dates in a centralised public database. Given these constraints, it was not a feasible metric to include.

School Score

Schools are a highly popular signal for house price prediction, with the majority of articles making use of the metric to some level [43, 18, 44, 35, 29, 42]. Due to this, it is deemed an important spatial metric and one to include within this dataset.

School data was extracted from Department for Education (DfE) “get information about schools” dataset constructor [6]. This provided the following useful information about schools:

- Open dates
- Close dates
- Phase of education
- Statutory low age
- Statutory high age
- Gender
- Admissions policy
- LSOA code

Initially, the “phase of education” was used to separate schools into primary, secondary, and sixth form education. However, upon observing the data it was discovered that independent schools can span multiple phases of education, and so were written in the dataset as “not applicable”. Due to this, low and high statutory age requirements were instead used to classify these phases, as follows:

- **Primary** ($\mathbb{I}_{P,i,T}$) : 1 if $Age_{Low} \leq 11$ else 0
- **Secondary** ($\mathbb{I}_{S,i,T}$) : 1 if $Age_{Low} \leq 16$ and $Age_{High} > 11$ else 0
- **Sixth Form** ($\mathbb{I}_{F,i,T}$) : 1 if $Age_{High} \geq 18$ else 0

Explanatory data analysis indicated that sixth form school proximity was a more significant driver of LSOA desirability for first-time buyers than primary and secondary education (See Fig C.2 in Appendix C). Due to this, a higher base weighting was assigned to the sixth form phase. This method of classification of the three phases of education allowed multi-phasic schools to be seen instead as multiple individual schools in the rankings, as they provide the same uses that these individual single-phase schools would.

After classifying the school phases and defining base weightings, multipliers (W) were introduced for selective admissions policies (e.g. Grammar schools) and single-sex status (unisex), to reflect the market premium often associated with these institutions. For any individual school i , the phase-specific weights were defined as:

$$\begin{aligned} P_{i,T} &= (1.5 \cdot \mathbb{I}_{P,i,T}) \times W_{U,i,T} \times W_{S,i,T} \\ S_{i,T} &= (1.0 \cdot \mathbb{I}_{S,i,T}) \times W_{U,i,T} \times W_{S,i,T} \\ F_{i,T} &= (1.0 \cdot \mathbb{I}_{F,i,T}) \times W_{U,i,T} \times W_{S,i,T} \end{aligned} \tag{4.5}$$

In the above equation, W_U was defined as 1.2 if the institution was unisex and 1 otherwise, and W_S was defined as 2 if the institution was selective and 1 otherwise.

Finally, the total school score for a target LSOA was calculated using a spatial decay gravity model, taking the target LSOA’s centroid position and observing all LSOA school scores within a 15km radius. Here, the decay exponent was varied by phase: primary schools received the harshest decay as they are

strictly local, while secondary and sixth forms were given weaker decays to reflect their wider catchment influences.

$$SchoolScore_{LSOA,T} = \sum_{i \in 15km} \left(\frac{P_{i,T}}{d_{iT}^3 + \epsilon} + \frac{S_{i,T}}{d_{iT}^2 + \epsilon} + \frac{F_{i,T}}{d_{iT}^{1.5} + \epsilon} \right) \quad (4.6)$$

In the above equation, d_{iT} represents the distance in kilometres to the neighbour's centroid position, and ϵ represents a smoothing constant (0.2) used to prevent infinite scores for schools located directly within the target LSOA. Alongside this final score, an improvement score based on previous years was constructed:

$$SchoolScore_{\Delta,LSOA,T} = SchoolScore_{LSOA,T} - SchoolScore_{LSOA,T-3} \quad (4.7)$$

University Score

Although university proximity is a less traditional metric, with fewer articles implementing it, its correlation with housing prices in an area has been established [35]. Identifying whether a LAD is primarily a student-based district provides a unique signal (studentification), with many student-market landlords often exhibiting distinct investment behaviours compared to traditional residential landlords as they calculate their yield based on an individual room basis rather than the whole house, often deeming these areas more profitable [34]. This highly competitive environment can decouple local house prices from broader market trends, making university influence an essential socio-economic indicator.

No pre-made dataset was found depicting all universities with their start date, end date and LSOA locations for their facilities. Due to this, a web-scraper was used to create an initial CSV file of all the main universities with their start and end dates. This was then manually altered as the strict definition of a university, an institution of higher education authorised to grant academic degrees, meant that some large institutions which would have an impact on house prices fell short of this definition. For example, UCL was founded in 1826, however, its academic degrees were awarded by the University of London up until 2023, where it was officially awarded university status. Situations like this meant some start dates required alteration.

After this, because universities are continually adding and removing buildings from their campuses, a LSOA level definition was deemed too messy. Therefore, I went one layer higher in abstraction and focused on the LADs where the primary campuses reside. The Nominatim API [20] was used to retrieve these LAD locations for each university.

Student populations were then added to the dataset through the Higher Education Statistics Agency (HESA) yearly student data tables [13]. Due to this being university specific rather than LAD specific universities that span multiple LADs divided this population figure equally across their districts (this is a potential source of inaccuracy as not all campuses are equal in size). From this a spatial decay score using LAD centroids was calculated as follows:

$$I_{LAD,T} = \sum_{i \in 30km} \frac{M_{i,T}}{(d_{iT} + 1)^2} \quad (4.8)$$

In the above equation, i represents an institution and $M_{i,T} = Population_{i,T} \times PrestigeWeight_{i,T}$. $PrestigeWeight$ is a metric for classifying the prestige of the university. The top ranked universities: Oxford, Cambridge, LSE, Imperial, UCL and KCL scored 3, all other Russell Group universities scored 2 and finally, any other university scored a 1.

Although Equation 4.8 shows the size and prestige of universities in each LAD it does not show the proportion of the city students make up. Due to this, the number of households per LAD was calculated using the government's live tables on dwellings [10] and the final score was created alongside an improvement score through its improvement over previous years:

$$UniScore_{LAD,T} = \ln \left(1 + \left(\frac{I_{LAD,T}}{Households_{LAD,T}} \times 100 \right) \right) \quad (4.9)$$

$$UniScore_{\Delta,LAD,T} = UniScore_{LAD,T} - UniScore_{LAD,T-3} \quad (4.10)$$

Business Score

Distance to the nearest central business district (CBD) is a primary predictor in many articles observing residential property values [43, 35]. With CBDs often exhibiting agglomeration effects, characterised by a dense concentration of high-order employment, retail, and cultural amenities, this often creates a level of prestige upon the area, increasing a LSOA's value. Overall, this makes determining business districts a key goal for the dataset.

As the UK does not have predefined business districts a formula was created to determine high business areas. First, the annual data on births, deaths and survival of businesses in each LAD was compiled. For this, the ONS provides archived demographics between 1995 and 2007 that determines the deaths and births of businesses through their VAT registrations and de-registrations [26]. For the demographics between 2010 and 2024, the ONS instead determines the birth and deaths through the Inter-Departmental Business Register (IDBR), a combination of VAT and PAYE (tax) records [24]. However, between these two periods there was a two year gap where the ONS was transitioning to the methodology. Here, there is only a 2009 region level report archived within the ONS [21].

$$Birth_{LAD,2009} = Birth_{LAD,2007} \cdot \frac{Birth_{Regional,2009}}{Birth_{Regional,2007}} \quad (4.11)$$

To fill this two year gap to a LAD level the 2009 regional statistics were taken and used alongside the 2007 regional statistics to determine a region wide multiplier in which the 2007 LAD level statistics could be multiplied, this is shown for the birth rates in Equation 4.11. This assumes all districts within a region grew consistently with previous years. Finally, 2008 data was created through linear interpolation between 2007 and 2009.

The number of households per LAD was determined using the government's live tables on dwellings [10] and a LAD level score was determined through a combination of business density and momentum:

$$\begin{aligned} LocalBusinessScore_{LAD,T} &= (\text{Norm}(\ln(Density)) + 1) \cdot (\text{Norm}(Momentum) + 1) \\ Density &= 1 + \frac{ActiveBusinesses_{LAD,T}}{Households_{LAD,T}} \\ Momentum &= \frac{Births_{LAD,T+1}}{Deaths_{LAD,T+1}} \end{aligned} \quad (4.12)$$

In the above equation, Norm describes min-max scaling ($\text{Norm}(x_{i,T}) = \frac{x_{i,T} - x_{min,T}}{x_{max,T} - x_{min,T}}$).

Finally, a spatial decay gravity model was introduced using the target and neighbouring LADs centroid positions to output a final score and an improvement score through its improvement over previous years:

$$BusinessScore_{LAD,T} = LocalBusinessScore_{LAD,T} + \sum_{i \in 20km} \left(\frac{LocalBusinessScore_{i,T}}{d_{iT}^2 + \epsilon} \right) \quad (4.13)$$

$$BusinessScore_{\Delta,LAD,T} = BusinessScore_{LAD,T} - BusinessScore_{LAD,T-3} \quad (4.14)$$

Transport Score

Transportation is a key indicator for gentrification [11], and so is a common metric for price evaluation in many articles [43, 18, 44, 35, 39, 42]. Coupled with a business score, this creates a strong metric for predicting desirable areas for many first-time buyers.

The National Public Transport Access Node (NaPTAN) dataset provided by the Department for Transport (DfT) [7] provides the following useful information about transport stops for England and Wales:

- Stop type (railway, metro, tram and bus)
- Longitude
- Latitude
- Creation date
- Modification date
- Status

One issue with this dataset is that it only begins at 2005. Due to this, I interpolate the initial 10 years via a historical backfilling heuristic, assuming anything constructed before 2005 was also present in 1995. A dataset can then be constructed by using the ONS' LSOA 2021 boundaries [22] and the longitude and latitude of the stops to determine which LSOA each stop belongs to.

$$V_{LSOA,T} = \sum_{i \in LSOA} (W_{StopType}) \forall \text{ stops where } StartYear \leq T \leq EndYear \quad (4.15)$$

Equation 4.15 describes a weighted volume where the weights are as follows: railway (5.0), metro (3.0), tram (3.0), bus (1.0). These weightings were determined through the value each transport facility provides. With this a hub multiplier was constructed:

$$V_{LSOA,T} = \begin{cases} 3 \cdot V_{LSOA,T} & \text{if } \sum(Stops_{LSOA,T}) > 5 \\ V_{LSOA,T} & \text{otherwise} \end{cases} \quad (4.16)$$

Finally, this weighted volume can be introduced into a spatial gravity decay model to construct both a transport score and an improvement score through its improvement over previous years.

$$TransportScore_{LSOA,T} = \ln \left(1 + \sum_{i \in 3km} \frac{V_{i,T}}{d_{iT}^2 + \epsilon} \right) \quad (4.17)$$

$$TransportScore_{\Delta,LSOA,T} = TransportScore_{LSOA,T} - TransportScore_{LSOA,T-3} \quad (4.18)$$

4.2.3 Spatio-Temporal Lag

As spatio-temporal lag is a high performing metric at the property-specific level [35, 18], this score was introduced to the dataset to observe its performance at a LSOA level. Adapted from the spatio-temporal lag feature discussed by Soltani et al. [35] (see Section 2.1.1). This score was used to analyse whether lag is still a useful feature when at the LSOA level rather than the individual building level. This adapted formula is as follows:

$$STLag_{LSOA,t} = \frac{\sum_{i \in 2km, j \in (t-6qtrs)} (W_{Spatial} \cdot W_{Temporal} \cdot G_{yoy,i,j})}{\sum_{i \in 2km, j \in (t-6qtrs)} (W_{Spatial} \cdot W_{Temporal})} \quad (4.19)$$

$$W_{Spatial} = \frac{1}{d_{it}^2 + \epsilon}$$

$$W_{Temporal} = e^{-\lambda j}$$

In the above, λ represents the temporal decay (0.1) and ϵ a smoothing constant (0.5) used to prevent infinite scores.

Two primary changes to the score were introduced: the first was the introduction of $W_{Temporal}$. In the original version, this lag took all sales within a period of the last x years and weighted them all equally, focusing primarily on the spatial distance of these sales. Instead, $W_{Temporal}$ adds a penalty to ensure older sales have a smaller impact on the overall score. The second change was the growth score, removing the original raw house prices as a nearby house in a more expensive area, such as Eton next to Slough, would skew the results. Instead, $G_{yoy,LSOA,t}$ (Equation 4.2) was used to move this price into the context of the increasing value of residential areas.

4.3 Model Construction

As briefly discussed in the methodology overview and dataset construction, shock years were removed along with the two test regions (North West and London), and the constructed dataset was then split (see Fig C.1 in Appendix C for the folder structure) into a dataset containing a three-year span of features and a dataset of ground truths for every LSOA. These datasets were then split into training (predictions for 2001 – 2018), validation (predictions for 2019 – 2021), and testing (predictions for 2022 – 2025).

Within this section, four regression models were constructed with the goal of predicting the percentage increase of LSOAs over a three-year horizon. The models, as described in Section 3, are as follows: MLR, GTWR [15], XGBoost [4], and Spacetimeformer [12]. These models were evaluated through observing and comparing the MSE and R^2 of each model for both a one-year and three-year prediction.

4.3.1 Multiple Linear Regression

MLR is the simplest solution to the problem of predicting house prices and has been used in many past articles as a basis for comparison [44, 35]. Due to this, it was used as a base model in which to compare the later, more sophisticated models.

This model begins by taking all the training year folders and concatenating them into a singular data-frame for training and another for ground truths ($Return_{LSOA,3yr_t}$). Each LAD was then one-hot encoded to capture the regional spatial variation of the data. LSOA codes could not be one-hot encoded due to their high cardinality (over 30,000 unique units) exceeding any viable RAM capacity, and were therefore excluded. Following feature scaling and the removal of non-numerical features (such as LSOA codes and LAD codes), the data was then passed through the MLR model. Here, the parameters were estimated using the Ordinary Least Squares (OLS) method. This provides a closed-form analytical solution that identifies the unique set of weights that minimise the sum of squared residuals. Unlike the XGBoost and Spacetimeformer models, this approach does not require hyper-parameter tuning, ensuring a deterministic and stable baseline for comparison against the non-linear models. The final model was then evaluated on the validation set to establish the baseline performance metrics as a reference for the tuning of the non-linear models.

4.3.2 GTWR

This model was selected as it is the most relevant form of regression model for this problem, accounting for both geographical and temporal characteristics. This is also supported by previous work, with the original GTWR article by Huang et al. [15] predicting the housing market for their primary example case, and many other papers using its predecessor, GWR [18, 43] for this problem.

In implementing GTWR I found the Python library was deprecated; due to this the model was instead developed in R. The primary issue with this is that R is not created for GPU use, with the `GWmodel` relying on CPU-bound cross-validation, an inherently slow and iterative process. To improve this, the code was parallelised to use multiple CPU cores to speed up training. Alongside this, the size of the dataset became an issue, with GTWR crashing after constructing a spatio-temporal distance matrix connecting all 33,755 LSOAs over the training set of 17 years, resulting in a massive RAM requirement of $2TB$ to hold this. To make this viable, the data was thinned, taking 10% of the LSOAs from each LAD to ensure local signals from all districts were still captured; this brought the RAM requirement down to $300GB$. To lower this further, temporal data from earlier years was removed, passing only the data used to predict the years 2013 to 2018, bringing the RAM into the more viable range of $45GB$.

With the setup complete, an adaptive bandwidth was found through taking a sample from the dataset, passing it through the model, and upscaling it to better represent the entire dataset through the use of sample size scaling (Equation 4.20). However, despite chunking, fitting the dataset to the model was still too computationally intense, with 1 of 107 chunks taking more than 3 hours to complete. Due to this, the model was deemed not suitable for a full evaluation of all LSOAs from England and Wales and was therefore not further explored as a viable pipeline. However, as first-time buyers typically only want predictions for a specific district, a model was run on the area of Bristol, using the Bristol Hippodrome as the centre point and predicting all LSOAs within a 10 mile radius of this point. Here, the model got a MSE of 0.1131 and a R^2 score of -0.9775 , suggesting that the extensive data filtering led to a model that struggled to capture the underlying market trend indicators, leading to a poor predictor.

$$b_{full} = b_{sample} \cdot \left(\frac{n_{full}}{n_{sample}} \right)^{\frac{4}{5}} \quad (4.20)$$

4.3.3 XGBoost

With many past articles concluding that gradient boosted trees consistently perform to a high standard in this problem space [44, 35, 42] (with Wang et al. [42] showing instances of their self-attention model losing to this baseline), XGBoost was a logical and sophisticated model to compare within this paper. The model's underlying logic, sequentially partitioning the feature space based on conditional thresholds, provides an intuitive framework for capturing the non-linear interactions inherent in the real estate markets. Furthermore, the integration of the SHAP (SHapley Additive exPlanations) library allows for a granular assessment of feature contributions, moving beyond the black-box conclusions just MSE and R^2 could provide and giving further insights into which indicators drive market appreciation and how.

Utilising the XGBoost library provided by Chen et al. [4], the training data was aggregated into unified data-frames. The LSOA centroids were then extracted and joined onto the dataset to provide the fundamental spatial signal, before explicitly pruning the LSOA and LAD codes alongside any other non-numerical features that would not work within the model. This data was then encapsulated into an optimised DMatrix structure for training. The model utilised a histogram-based tree method to accelerate boosting, minimising the RMSE. With the use of the validation set, hyper-parameters and early stopping were then tuned to prevent over-fitting.

4.3.4 Spacetimeformer

During research there was very little detailing the use of more modern solutions within real-estate predictions, such as the use of transformer models. An article by Wang et al. [42] confirmed the plausibility of a transformer method: solving a similar estate prediction problem using satellite imaging along with local facility data to determine an individual property’s price. This paper aims to implement and compare the “state-of-the-art” Spacetimeformer model, found in the parallel geospatial and temporal problem fields of weather and traffic forecasting [12].

To ensure the Spacetimeformer model works with the long-range temporal signals of the real-estate market the model had to undergo some alterations. The data-formatter `CSVTimeSeries` underwent the majority of alterations. First replacing the traditional time features: “Yearly”, “Monthly”, “Weekly”, “Daily” and “Hourly”, with just “Yearly” and “Quarterly” to reflect the format of this paper’s dataset (this was also updated within `time.features`). Next, `__len__` and `__getitem__` were implemented to enable efficient batching in the training loop. The assumption that each feature has its own time encoding was removed, aligning all features to use the same quarterly timestamp. Finally, it was simplified to not automatically split data into train, validation and test sets as my own data cleaning phase already handled this.

With this setup complete, the model was then trained as follows. First the data was aggregated into a unified dataset and the centroids for each LSOA were found and joined. Next, this dataset was passed into the `CSVTimeSeries` and encapsulated into a data loader, so training data could be passed into the model via batches. The Spacetimeformer architecture was then established by setting both the global and local self attention to “performer”. This architectural choice allowed the model to learn long-range temporal dependencies and complex spatial correlations whilst still running within a viable time-frame. Due to the size and complexity of this model the data loader used multiple CPUs and the model itself was run off a GPU on Bristol’s BluePebble supercomputer. To reduce over-fitting and speed up the run time, a subsample of 75% of the data was taken for training. Early stopping was also implemented to minimise over-fitting.

Chapter 5

Classifier Construction

5.1 Outline

This section presents the move from regression predictions into classification. Classifying each LSOA based on the percentage increase over a given three-year period, the model aims to distinguish the top 25% of performing areas. The sections of this chapter have been ordered to first describe the modifications to the dataset and then the implementation of the staged classifiers.

5.1.1 Methodology Overview

Past articles have argued that a regress-then-classify approach has two primary benefits over a direct classification approach, despite performing at a similar level. First, a regress-then-classify approach can lead to superior flexibility due to its ability to alter classification boundaries rather than fully retraining a model. Second, it moves the model from a black-box to a grey-box approach, providing the option for experts to evaluate the underlying regression predictions alongside the final classification to make a more informed decision [3, 36].

Due to the benefits of a staged approach, the four regression models are passed through an XGBoost classifier, training the classifier based on the regression models' training set predictions and classification boundaries to produce these final predictions (this allows the classifier to learn the regression model's biases and patterns and adapt accordingly). Alongside this, a direct classifier has also been constructed for MLR and XGBoost for comparison. Due to the resource-intensive nature of the Spacetimeformer it was decided to only construct the regress-then-classify pipeline for this model for the reasons discussed above.

Shifting the underlying problem from regression to classification has facilitated the use of additional evaluation metrics. The following models will be evaluated based on their weighted accuracy, recall, and F1-scores, alongside their confusion matrices.

5.1.2 Boundary Justification

With the majority of housing areas typically increasing in value over time, the raw output of percentage increase provided by regression models has limited interpretability, missing the relative context of how the area performs in contrast to the rest of England and Wales. To gain this context and the ability to learn a specific model's bias, a shift to a classifier model was chosen over a simple comparison of an area's percentage increases. With past articles stopping at a regression output there is limited justification for a set optimal boundary. To determine a valuable boundary that is still embedded within a financial context, various significance levels were explored and the percentage increases were compared to those of the S&P 500 [19] and FTSE 100 [5] (after stamp duty), allowing the financial alternative of traditional indices to be compared in order to justify the chosen boundary. After the analysis, it was found that a top quartile boundary performed at a similar level to these traditional indices, as shown in Fig 5.1.

This boundary was further validated by institutional standards, where it was found that Cambridge Associates (a leading global investment firm for institutional investors) use the top 25% as the threshold for identifying institutional-grade, high-performing funds, effectively aligning professional investment benchmarks with the model's classification boundary.

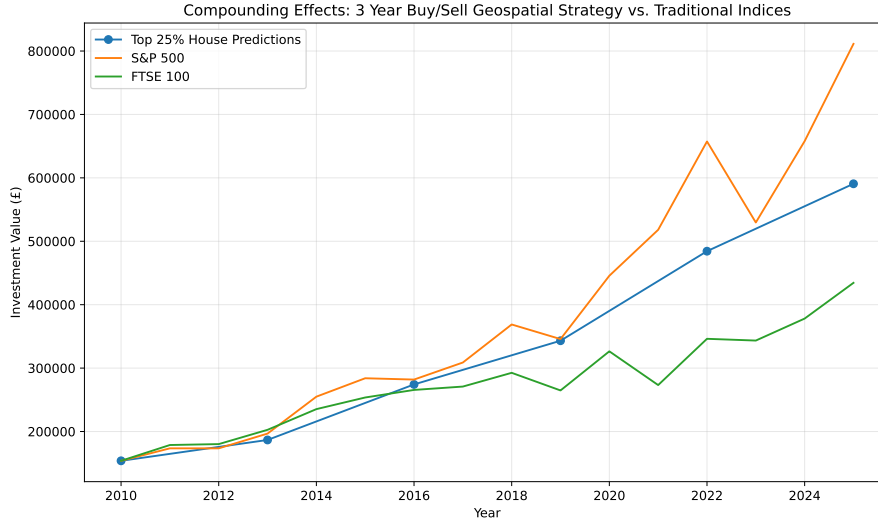


Figure 5.1: A comparison of the compounding effects of various stock markets and buying a house within the top 25% of LSOA areas every three years between the years of 2010 and 2015

5.2 Dataset Modification

To facilitate the binary classification score, the dataset was augmented with a performance label derived from the the HM Land Registry Price Paid Data (PPD) [14]. Here, a dynamic yearly boundary was established based on the upper quartile (75th percentile) of all LSOA $Return_{LSOA,3yr_t}$ scores for a given prediction point t . This relative threshold ensured that the model could correctly identify the outperforming LSOAs within the specific economic context of that year. Areas exceeding this dynamic boundary were assigned a value of 1, representing high-growth “investment targets” for the buyer persona, while the remainder were labelled 0.

5.3 Model Construction

5.3.1 Regress-Then-Classify

The construction of the staged regressor-classifier models followed a cascaded architecture designed to leverage the high-capacity feature extraction of the regression models. The training data was aggregated and formatted consistent with the requirements for the chosen regression model in Section 4.3. Next, the regression model was constructed and its trained weights loaded in. The aggregated data-frame was then passed through the regression model as training data and its $Return_{LSOA,3yr_t}$ predictions were compiled into a DMatrix and passed into the XGBoost model, alongside the ground truths of aggregated classification scores. This model utilised a histogram-based tree method with a binary logistic objective. To account for the inherent class imbalance of the 75th percentile threshold (where only 25% of the data represents the positive signal), a `scale_pos_weight` of 4.0 was applied. This ensured the boosting process penalised the misclassification of high-growth more heavily, prioritising recall. Performance was monitored via the Area Under the ROC Curve (AUC) to assess the model’s ability to distinguish between classes independently of the classification threshold.

5.3.2 Direct Classification

Despite the benefits of the regress-then-classify pipeline, as given in the methodology overview, a staged pipeline also has disadvantages. First, direct classifiers are inherently more robust to residual noise and outliers; by collapsing the continuous price distribution into a binary objective, the model ignores extreme price variances that do not cross the decision boundary. Second, having a singular phase means the model can optimise for the given boundary, avoiding error propagation by refining weights specifically for the given decision boundary, potentially yielding superior class separation in high-volatility environments.

Logistic Regression

To adapt the MLR into a classifier model (logistic regression), the architecture was altered so the results of the forward pass were passed through a sigmoid function ($\phi(x) = \frac{1}{1+e^{-x}}$), restricting the continuous output into a probability which given a threshold (0.5) can produce a binary classification. For training, a similar pipeline to the regression variant was taken. Training data was aggregated into a singular dataset and the LADs were one-hot encoded before all non-numerical features were filtered out. In this configuration, the continuous $Return_{LSOA,3yr_t}$ targets were discarded in favour of the binary classification labels derived in Section 5.2. Features were then passed into the logistic regression model.

XGBoost

For the construction of the XGBoost direct classifier, the development followed a similar pipeline to the regression variant. Training data was aggregated into a singular dataset and the LSOA centroids were joined to provide the fundamental spatial signal before filtering out all non-numerical features (such as LAD codes). In this configuration, the continuous $Return_{LSOA,3yr_t}$ targets were discarded in favour of the binary classification labels derived in Section 5.2. Features were then encapsulated into an optimised DMatrix structure and passed into the XGBoost model. This model used the same methods and underlying architecture as the regress-then-classify models described in Section 5.3.1.

Chapter 6

Model Development and Architectural Optimisation

6.1 Outline

This section presents the model optimisations that occurred through feature analysis on the validation set and a comparison of classifier architectures on the test data. The sections of this chapter have been ordered to first present an ablation study of each feature then, removing poor features, present an analysis of regress-then-classify models and direct classifier models.

6.2 Feature Optimisation

To evaluate the relative contribution of each feature (a complete description of features can be found in Table C.1 of Appendix C), a Leave-One-Out Ablation Study was conducted. Rather than assessing features in isolation, which fails to account for feature interactions, this study systematically removed individual features from the full model to measure the subsequent decrease in predictive power. XGBoost was utilized due to its ability to handle non-linear relationships (unlike MLR and GTWR), whilst training in a reasonable time-frame for this extensive study (unlike the transformer).

Observing the marginal increase in R^2 of the regression model on the validation set (this was done on the validation set rather than the test set to ensure the models didn't cheat) after the removal of specific variables, the power of each signal was quantified. This process allows for the identification of critical features (those whose removal causes significant performance loss) versus redundant features (those whose information is already captured by other variables). The results of this sensitivity analysis are visualized in Fig 6.1a.

Following the initial assessment of the base model's feature power, a Recursive Feature Elimination (RFE) process was conducted to optimise the predictive signal. In each iteration, the variable with the most detrimental impact on R^2 was pruned, followed by a subsequent leave-one-out analysis. This cycle was repeated until the feature set reached a state where all remaining features exhibited positive feature power. This final optimised subset is shown in Fig 6.1b, where both "school score" and "business score delta" were removed. Here, the LAD Price Index ($LPI_{LSOA,t}$) remained the strongest signal for the validation set.

An RFE was chosen over the simpler method of removing all detrimental features from the base model for two primary reasons. First, the removal of the spatial noise caused by the "business score delta" altered the importance landscape. This was evidenced by the "transport score", where initially negative was found to be positive in subsequent RFE iterations. The second issue with this method was feature dependencies; it was found that removing "transport score" from the base model alongside "business score delta" had a large negative effect on the overall model due to the "business score" depending on it.

With the optimal feature subset established, the following section evaluates and compares the Direct Classifier and Regress-Then-Classify pipeline, observing the predictive performance in terms of classification abilities.

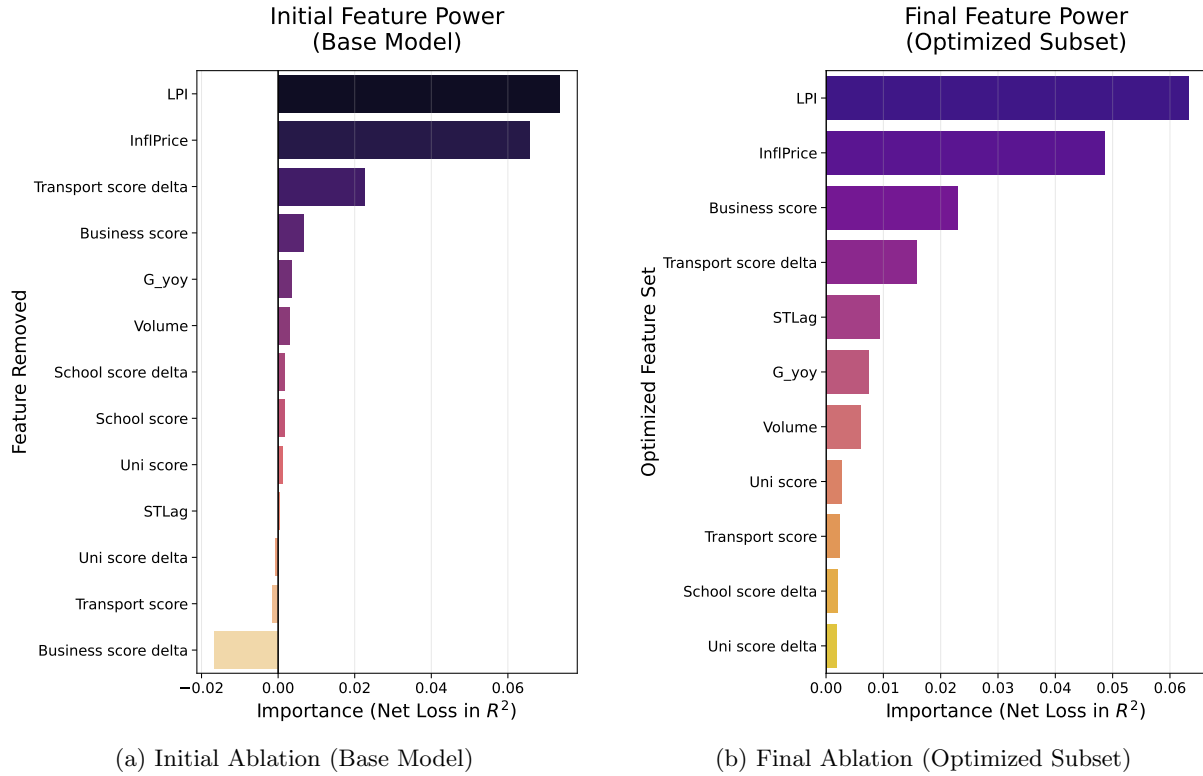


Figure 6.1: Feature Power Comparison: Showing the transition from the high-dimensional base model to the optimized feature subset via Recursive Feature Elimination (RFE).

6.3 Architectural Strategy (Regress-Then-Classify vs. Direct)

It is computationally feasible to train both a regression model and a separate classifier model for MLR (logistic regression for classification) and XGBoost models. The combined training time for separate regression and classification models remains substantially lower than that of a single transformer-based regression model. Evaluation for both strategies took place on the test set to determine the optimal methodology for the two architectures. Fig 6.2 shows an initial high-level comparison of the strategies at various data states. Here, various data states were used to observe how potential residual noise, introduced by spatial features and STLag, could impact a regress-then-classify pipeline.

6.3.1 MLR with XGBoost Classifier vs. Logistic Regression

Observing Fig 6.2 it is shown that a multiple linear regressor passed into an XGBoost classifier resulted in a higher F1-score than a logistic regressor. This is due to the XGBoost classifier’s ability to capture non-linearity, learning from the poor regression predictions (later shown in Table 7.1 and Table 7.2 of Chapter 7) and mapping it into the non-linear context required for better classification.

While the F1-score remained constant for the MLR regress-then-classify pipeline across all feature introductions, spatial features increased the model’s recall (0.65) at the expense of precision (0.34). This increased recall upon the addition of spatial is due to the new signals allowing the model to understand an area’s attributes over purely its financial details, thereby leading to the identification of more top areas. Afterwards, the introduction of STLag stabilised the model, bringing it to a middle ground between precision (0.35) and recall (0.63), with the greater dynamic momentum of this signal (compared to the static nature of spatial attributes) allowing it to better contextualise the growth of a potential area and determine whether it had ended.

The logistic regression model began with a much higher precision (0.63) and much lower recall (0.25), implying this direct pipeline is more risk averse in its predictions, with no penalisation for high-growth misclassification (such as XGBoost’s `scale_pos_weight`), leading to a model that plays safe and predicts 0 if in doubt (this is shown in Fig 6.3). With the introduction of spatial features there was an increase in precision (0.65) at the expense of recall (0.22); with this lack of high-growth penalisation creating a model which uses this spatial information to create a stronger bottom 75% LSOA classifier. Finally,

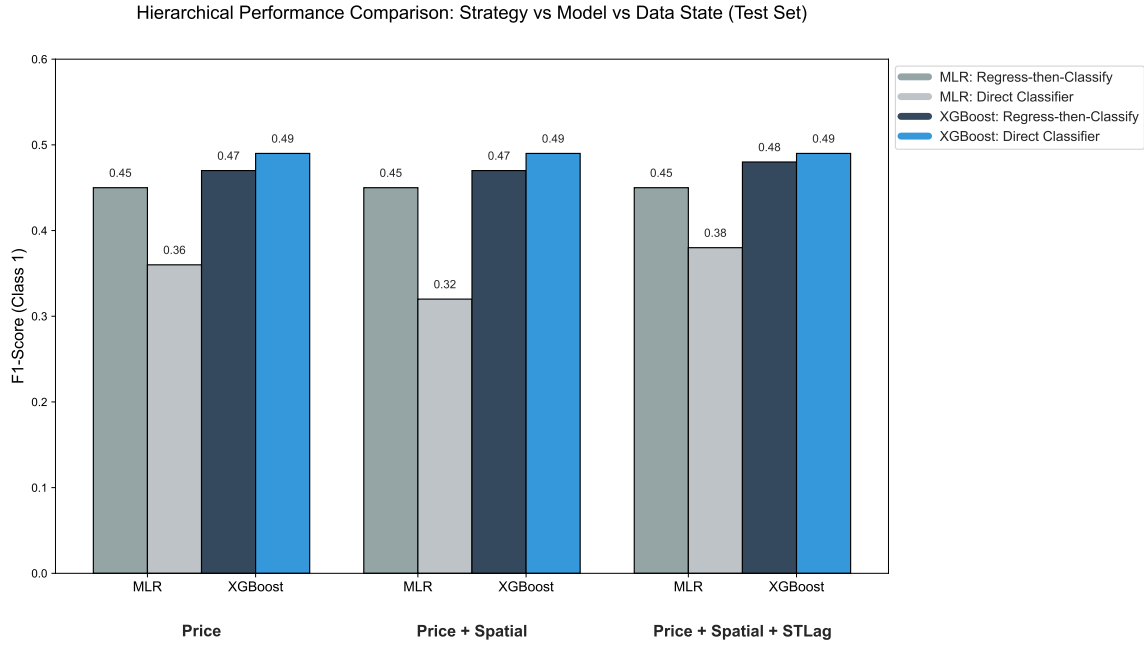


Figure 6.2: Comparison of the Regress-Then-Classify and Direct Classifier pipeline on the test set

the introduction of STLag, a feature inherently linked to an area’s potential value growth (unlike spatial features which are linked to an area’s attractive attributes) forced an increase in recall (0.28) and a subsequent decrease in precision (0.61).

The gap in F1-scores between the two strategies, shown in Fig 6.2, increased to a gap of 0.13 with the introduction of spatial data and subsequently decreased to a gap of 0.07 with the introduction of STLag. This shows that although a direct approach can be more robust, in the case of an XGBoost classifier vs. logistic regression, the benefits of non-linear classifications and the ability to penalise high-growth classification ultimately outweigh the direct pipeline benefits, creating a better model for class 1 predictions over a greater general model. This is shown in Fig 6.3 where although the direct pipeline is a less balanced model, it represents a stronger overall general model due to it accurately predicting 96.37% of the much larger bottom 75% class. This results in a much greater global accuracy than the balanced 68.38% of the regress-then-classify model. This figure also shows the greater potential of the regress-then-classify model with it identifying just under 3 times more of the total top quartile class. However, this is at the cost of precision, something the first-time buyer persona will likely prioritise.

In the subsequent sections a greater F1-score will be prioritised over precision to create the best general framework for classifying the top 25% of LSOAs. The advantages of models that prioritise each metric (F1-score, precision and recall) will be analysed in Chapter 7, where architectural adjustments will be suggested to align this baseline model with the specific risk profile of a first-time buyer. Therefore, in the subsequent chapters a MLR, regress-then-classify pipeline, will be used over logistic regression.

6.3.2 XGBoost Regress-Then-Classify vs. Direct Classifier

Fig 6.2 shows that a direct classifier marginally outperforms a regress-then-classify pipeline. This is due to the reasons listed in Chapter 5, specifically, the model’s greater robustness to outliers caused by the simpler binary objective and its the model’s ability to optimise for the given boundary, avoiding error propagation through refining weights specifically for that boundary.

Observing Fig 6.2 a consistent F1-score of 0.49 occurs for all data states for the direct classification model. However, unlike findings in Section 6.3.1, the precision and recall remained constant in all states (other than the recall for pure financial data which sat at 0.67 rather than 0.66), implying that it can extract a known area’s potential based mainly on financial data alone. However, observing the model’s ability to make predictions on the unknown regions of London and the North West the direct and regress-then-classify pipelines both saw variation in precision and recall with the introduction of features. Here, the recall for both improved with spatial features at the expense of precision before stabilising at a middle ground for both precision and recall in the subsequent STLag introduction. This implies that although

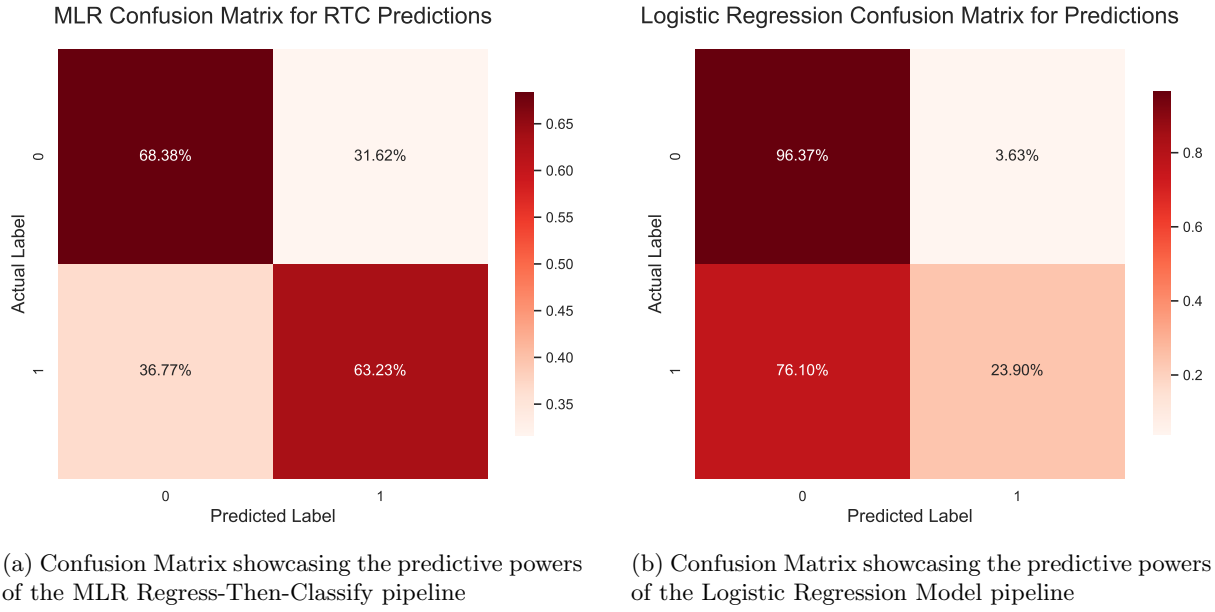


Figure 6.3: Pipeline Predictive Power Comparison: Showing the predictive abilities of both pipelines predicting 3 years into the future using the optimised subset of features (see Section 6.2)

the model is powerful enough to predict known regions with just financial data it is still learning the trends of spatial data and STLag.

The test set showed that the regress-then-classify model's scores changed with the introduction of extra features, where the recall increased with spatial data whilst the precision decreased. The model ultimately stabilised on a middle ground with the introduction of STLag, leading to its top F1-score. This variance is consistent with the idea that a regression portion of the pipeline reacts to outliers, making the model less robust overall.

The reaction to additional features from the three models using XGBoost in their pipeline, as shown in Fig 6.2, implies that although the regressor component of the classifier can have an impact on classification the primary determinant is the classifier in which it is passed into.

Observing the confusion matrices within Fig 6.4, results show that although the two models are similar in outcomes, the direct classifier outperforms the regress-then-classify model in identifying the top 25% of LSOAs, with a recall of 0.66 compared to 0.58. However, this comes with a slight decrease in precision, 0.39 compared to 0.41. With the similarity in matrices and the computational feasibility meaning the time to train a separate regressor and classifier has little practical impact the subsequent chapters will use a direct classifier. As calculated in Appendix C, the direct pipeline increases the potential search space for a buyer by 21%. Although this introduces a 25% increase in false-positive identifications, it successfully captures 15% more true-positive high-growth areas. For the first-time buyer, this is a trade-off to be discussed in Chapter 7 as it increases the buyer's opportunity to find a good, affordable house as they increase their search radius of potential top areas.

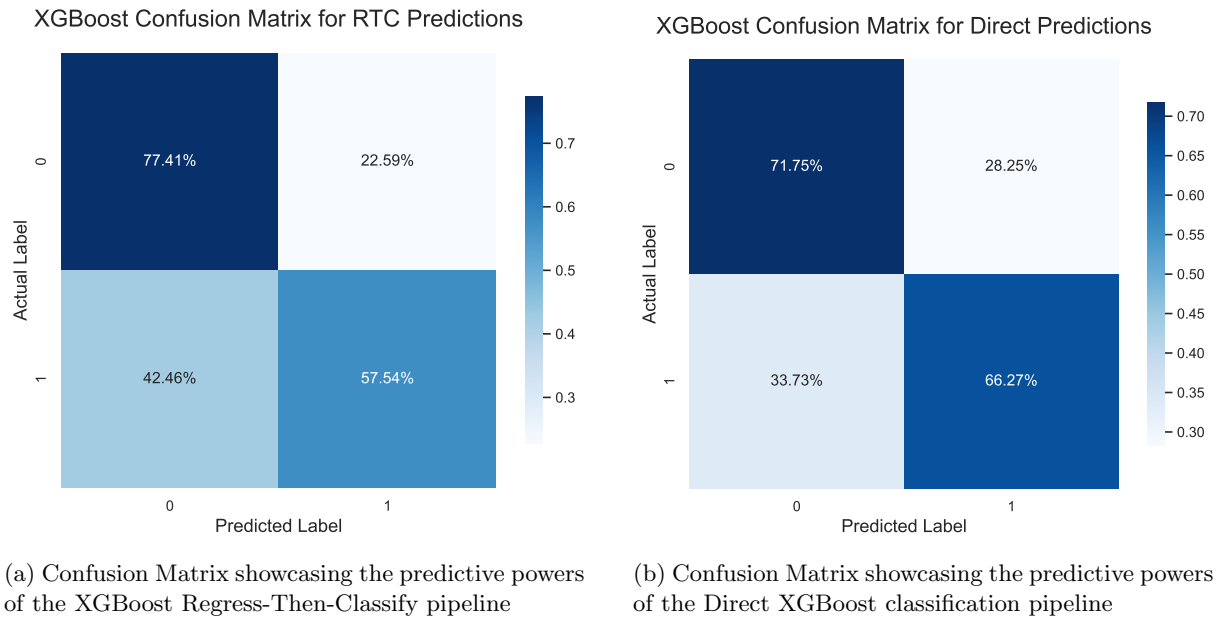


Figure 6.4: Pipeline Predictive Power Comparison: Showing the predictive abilities of both pipelines predicting 3 years into the future using the optimised subset of features (see Section 6.2)

Chapter 7

Empirical Analysis

7.1 Outline

This section presents the findings of both the regression models from Section 4 and the subsequent classifiers from Section 5. The sections of this chapter have been ordered to first present the results for each model, then further analyse its predictive powers on the areas of Bristol and London. The features will then be analysed using a SHAP graph to determine the cause of each feature's impact. Finally, a discussion on how to adapt the top model to better suit different personas is presented.

7.2 Evaluation of Model Performance

7.2.1 Results

Table 7.1 shows the performance of each regression model. Here we see that XGBoost outperforms the other two models in both MSE and R^2 score, implying it is the top architecture for regression. A closer analysis shows that an XGBoost model using price and spatial features is the top performing model with an MSE of 0.0774 and R^2 score of 0.1315. This trend was previously observed in Chapter 6, where the regress-then-classify model saw an increase in recall with the addition of spatial information. This implies that spatial data is necessary for a strong prediction and at a LSOA level, STLag adds little value to the prediction, usually resulting in the over-saturation of financial data and becoming a detriment in all cases except MLR.

The R^2 scores presented in Table 7.1 showcase the complexity of the forecasting problem, with both MLR and Spacetimeformer having a negative R^2 , meaning the models performed worse than a horizontal line representing the mean of the observed data. While a value of 0 indicates a model that predicts no better than the mean, a negative R^2 signifies that the model's residual sum of squares is larger than the total sum of squares. This indicates that the linear assumptions of the MLR model and the simplified architecture of the Spacetimeformer were sufficiently misaligned with the underlying trends of the test set, likely due to Covid-19 altering market priorities and therefore feature importance between the years of the training data and the test data. This shift ultimately led to their predictions being further from the true values than a simple historical average would have been.

Table 7.1 and Fig 7.1 show that the MLR model predicts areas better than the Spacetimeformer model, as shown by its consistently lower MSE, and mean prediction in comparison to the true mean of the test set. However, whilst the MLR model was able to better capture the value of areas, it did so through taking a smoother, more global approach, resulting in a worse R^2 and smaller variance. This is likely why the model performed best with only financial data as the introduction of spatial features to the global focused model increased residual noise rather than localising predictions, due to the model lacking the ability to represent the non-linear spatial data.

The transformer model possessed the ability to learn the complex non-linear trends inherent in real estate, as reflected through its better R^2 score and greater variation represented. However, the computationally expensive long-range transformer model required simplification in order to be trained within a viable time-frame, resulting in a model with fewer hidden layers, reducing the model's ability to capture more complex trends, fewer encoders, simplifying the relationship between time and space, and fewer attention heads, meaning fewer matrices to learn specific key signals. This, coupled with the fact that

transformer models required a lot of data, resulted in a model that could not fully capture the real estate signals at a level comparable to XGBoost.

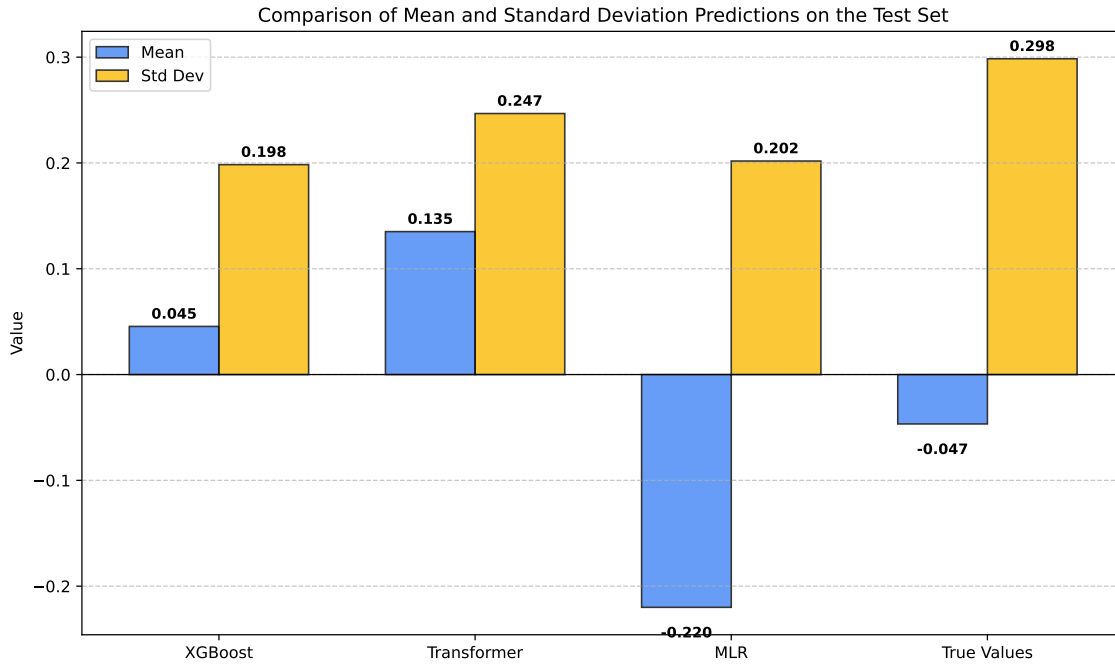


Figure 7.1: Comparison of all the models’ mean and standard deviations predictions on test set (predicting years 2022 - 2025). This uses the top feature combination of each model, as determined in Table 7.1.

Observing the one-year predictions provided in Table 7.1, it is shown that predicting area values over a shorter time-frame is a simpler problem with the MSE for all models being lower, whilst the R^2 values all increased. XGBoost provides an interesting result here as, unlike the three-year predictions where STLag introduced noise, causing a decrease in performance, the one-year prediction was less impacted by STLag’s inclusion, implying that although it is not an optimal signal to include for LSOA level predictions it does demonstrate a greater short term impact. This makes sense as after a three-year period the effects of this lag have already taken place and a new lag pattern formed.

Testing the performance of the regression models on the unknown regions of London and the North West (Table 7.2), the MLR and XGBoost models showed a decrease in overall performance, with greater MSE values and lower R^2 scores, implying they could not fully learn the estate trends and instead learnt what was typical for the regions. However, the Spacetimeformer model saw little change in performance between known and unknown regions, meaning this model could pick up the signals that imply an area’s future growth. This showcases the potential of a transformer model in real estate predictions as, although being computationally expensive, leading to a simplified architecture in this paper and therefore a lower performing model, it is a highly robust model in comparison to alternative methodologies.

Comparing the classifiers’ abilities on the test set, Table 7.1 presents evidence of the XGBoost direct classifier outperforming the other classifiers, with a greater F1-score (0.49) and precision (0.39). Observing the various feature combinations, little change in performance was found with a purely financial dataset slightly outperforming the other two configurations with a recall of 0.67. This implies that although spatial data likely adds value, the value lost through saturating the dataset is proportionate, and therefore a purely financial dataset is optimal for classification with an XGBoost model over a three-year horizon.

The weakest classification pipeline is the Spacetimeformer regression model with an XGBoost classifier. This model was extremely sensitive compared to the other two models, with a `scale_pos_weight` of 4 causing everything to be classified as in the top 25%, and a `scale_pos_weight` of 3 causing everything to be classified as in the bottom 75%. Due to this, limited tuning could occur, resulting in a classifier that displayed more aggressive behaviour, that unlike the other two models, could not be tuned.

Looking at classifications for a one-year horizon, F1-scores for all models decreased, with each model displaying more aggressive prediction behaviour, shown by a higher recall and lower precision. This is likely due to the classifiers being tuned for a three-year horizon alongside the fact that the increase over a one-year horizon is smaller, meaning less variance in value increase across all areas. This smaller variance

makes it harder to classify at a set boundary.

Testing on the unknown regions of London and the North West, MLR became the top model, with the other two models displaying more aggressive predictions with Spacetimeformer achieving a recall of 0.94. This is likely due to London being a sub-market that behaves differently compared to the rest of England and Wales, leading to the over-classification of top performing areas. While the MLR model also improved its recall, it maintained a consistent level of precision. This resilience is likely due to the linear regression component of the pipeline providing a form of implicit regularisation, making it less susceptible to over-fitting the unique signal variances of the London market.

Based on this analysis, in the subsequent sections the top performing classifier will refer to the XGBoost direct classifier using only financial features, and the top performing regression model will refer to the XGBoost regression model with both financial and spatial features.

Table 7.1: Model Performance Comparison across 1-year and 3-year Forecasting Horizons on the test data (predicting years 2022 – 2025)

Model	Features	1yr Reg.		3yr Reg.		Classifier - 1yr (Class 1)			Classifier - 3yr (Class 1)		
		MSE	R^2	MSE	R^2	Pr.	Re.	F1	Pr.	Re.	F1
MLR	Price	0.0949	0.1647	0.1296	-0.4546	0.17	0.73	0.28	<u>0.39</u>	0.54	0.45
	+ Spatial	0.0952	0.1626	0.1330	-0.4929	0.17	0.76	0.27	0.34	0.65	0.45
	+ Spatial + STLag	0.0953	0.1611	0.1329	-0.4920	<u>0.19</u>	0.67	<u>0.29</u>	0.35	0.63	0.45
XGBoost	Price	0.0830	0.2699	0.0781	0.1233	0.10	0.86	0.18	<u>0.39</u>	0.67	<u>0.49</u>
	+ Spatial	<u>0.0815</u>	<u>0.2829</u>	<u>0.0774</u>	<u>0.1315</u>	0.10	0.87	0.18	<u>0.39</u>	0.66	<u>0.49</u>
	+ Spatial + STLag	0.0820	0.2781	0.0802	0.0997	0.13	0.76	0.23	<u>0.39</u>	0.66	<u>0.49</u>
Transformer	Price	0.1602	-0.0723	0.3862	-0.3227	0.08	0.94	0.15	0.22	0.56	0.31
	+ Spatial	0.1584	-0.0663	0.3498	-0.2682	0.08	0.96	0.15	0.22	0.66	0.32
	+ Spatial + STLag	0.1562	-0.0794	0.3742	-0.3598	0.08	<u>0.97</u>	0.16	0.22	<u>0.79</u>	0.34

Note: Precision (Pr.) and Recall (Re.) are abbreviated for space.

Table 7.2: Model Performance Comparison across 1-year and 3-year Forecasting Horizons on the test region data (predicting London and the North West for the years 2022 – 2025)

Model	Features	1yr Reg.		3yr Reg.		Classifier - 1yr (Class 1)			Classifier - 3yr (Class 1)		
		MSE	R^2	MSE	R^2	Pr.	Re.	F1	Pr.	Re.	F1
MLR	Price	0.1267	0.0568	0.2442	-1.3442	0.16	0.56	0.25	<u>0.39</u>	0.62	<u>0.48</u>
	+ Spatial	0.1249	0.0702	0.2144	-1.0583	<u>0.17</u>	0.61	<u>0.26</u>	0.36	0.66	0.47
	+ Spatial + STLag	0.1482	-0.1034	0.2043	-0.9615	0.16	0.57	0.25	0.37	0.65	0.47
XGBoost	Price	0.1015	0.2447	<u>0.1295</u>	<u>-0.2436</u>	0.09	0.89	0.17	0.31	0.77	0.45
	+ Spatial	<u>0.1004</u>	<u>0.2527</u>	0.1314	-0.2614	0.12	0.81	0.22	0.28	0.82	0.42
	+ Spatial + STLag	0.1020	0.2410	0.1317	-0.2648	0.12	0.82	0.21	0.30	0.80	0.44
Transformer	Price	0.1723	-0.1054	0.4115	-0.4491	0.07	0.67	0.13	0.23	0.74	0.35
	+ Spatial	0.1512	-0.0539	0.3345	-0.2468	0.09	<u>0.97</u>	0.16	0.24	0.64	0.35
	+ Spatial + STLag	0.1483	-0.0610	0.3452	-0.2896	0.06	0.38	0.10	0.20	<u>0.94</u>	0.33

Note: Precision (Pr.) and Recall (Re.) are abbreviated for space.

7.2.2 City Prediction Analysis

To further evaluate model performance, spatial confusion matrices were constructed for both Bristol (test set) and London (unknown region set), predicting for the year 2022 with the top test set model (the direct XGBoost classifier). The winning model for the unknown region was not selected for this comparison to ensure a greater evaluation could take place on the top performing test set model, as a first-time buyer will likely not use a model that has not been trained on their specific region and so a deeper evaluation of its performance is ultimately less important.

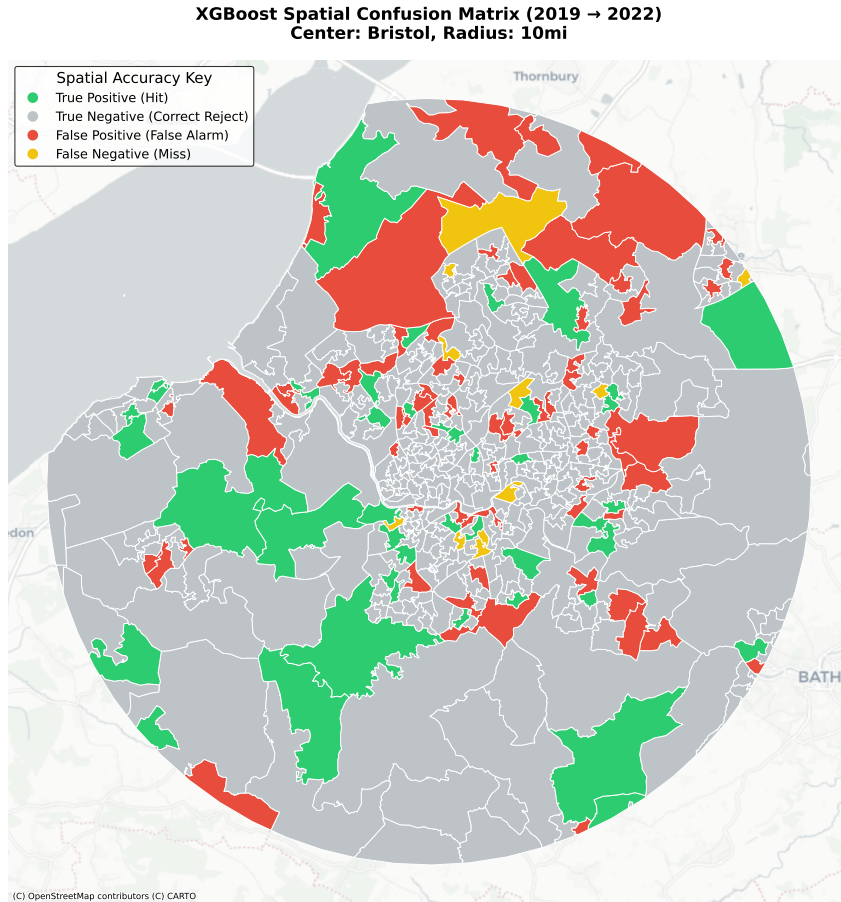


Figure 7.2: Spatial confusion matrix of the top performing classifier (XGBoost) for the test year 2019 (predicting 2022)

Fig 7.2 shows the spatial confusion matrix for Bristol, centring on the Bristol Hippodrome and covering a 10 mile radius. Here we can see that the model correctly identified 43 top 25% areas, missed 11 areas, falsely identified 70 and correctly rejected 401 areas, meaning it had a recall of 0.80, a precision of 0.38 and an F1-score of 0.52. These metrics are slightly above that of the general test set shown in Table 7.1 in both recall (0.66) and F1-score (0.49), however slightly worse in precision (0.39). This overall similarity to the test set justifies Bristol as a fair example of the model’s capabilities upon known data.

In Fig 7.2 we can see that the majority of misses come from smaller LSOAs. This is likely due to the differing impact of spatial decay on densely populated areas (smaller LSOAs) compared to sparsely populated areas (larger LSOAs). For example, the decay shown in transport score (Equation 4.17) spans a 3km radius. In a densely populated area there are likely more means of transport within a given 3km radius and therefore more noise present in the score. For example, a bus stop that is 2km away is more useful in an area where there is not a bus stop 500m away. Therefore, the signal from the 2km bus is less relevant for a first-time buyer looking at purchasing in a densely populated area. Splitting this transport score to a LSOA wide signal will somewhat combat this issue, as there is more area within the sparse LSOA in which the transport facilities may reside. However, residents living on the edge of an area will still have a score based on the LSOA’s centroid. Inversely, in a densely populated area, travelling a mile is more challenging than in a sparse area due to a greater population meaning greater traffic congestion.

Therefore, spatial decay with a set radius limits the potential power of a score, potentially explaining the false negatives and false positives found in smaller LSOAs.

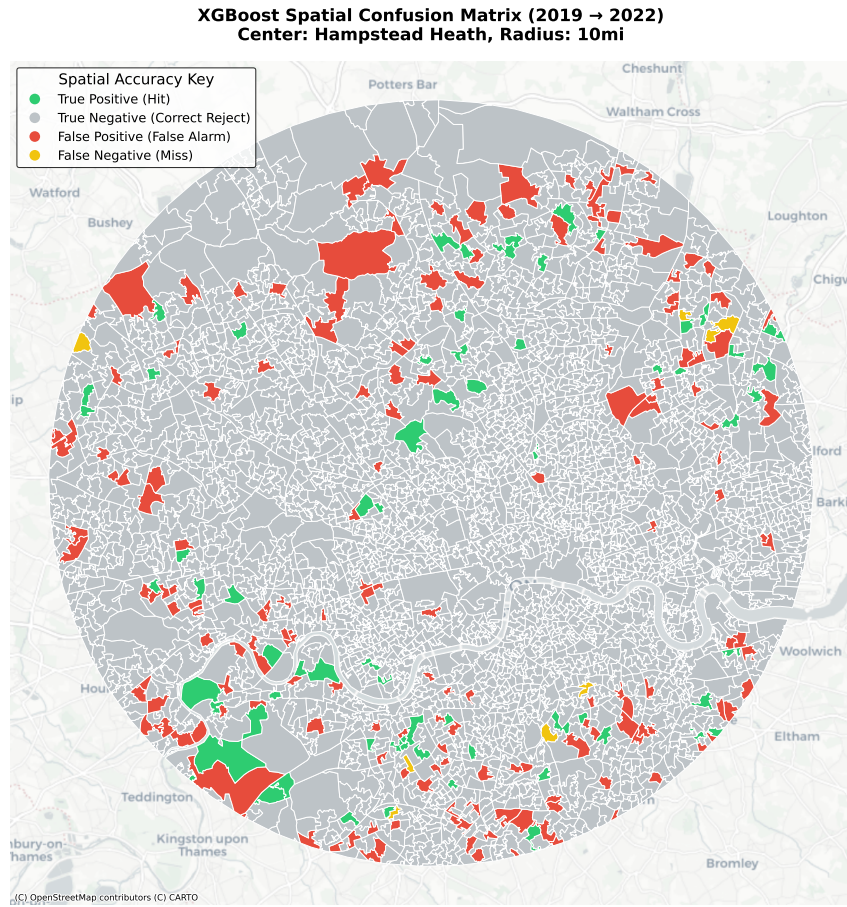


Figure 7.3: Spatial confusion matrix of the XGBoost classifier for the unseen region of London for year 2019 (predicting 2022)

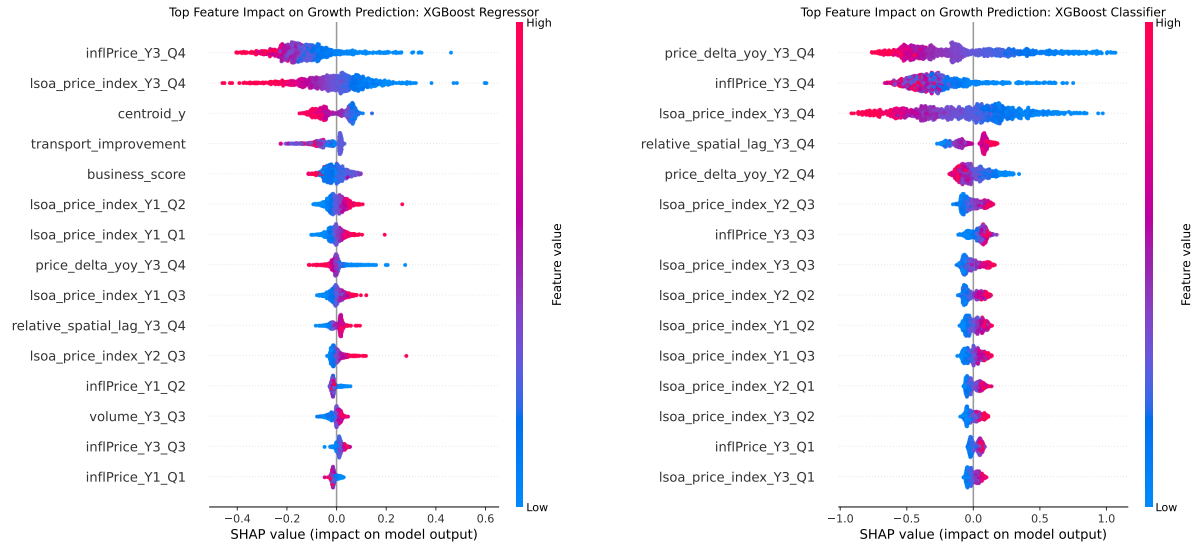
A confusion matrix of the performance of the XGBoost model in the unknown region of London can be seen in Fig 7.3. Here the model correctly identified 70 top 25% areas, missed 8 areas, falsely identified 175 and correctly rejected 3190 areas, meaning it had a recall of 0.90, a precision of 0.29 and an F1-score of 0.44. These metrics are representative of the general unknown region set shown in Table 7.2 with a greater recall (0.80), slightly worse precision (0.30), and the same F1-score (0.44). Due to this, the example of a 10 mile radius of Hampstead Heath is a fair visualisation of the model's performance in unknown regions.

Observing Fig 7.3, we see that it is a much more densely populated area than that of Bristol, shown in Fig 7.2. Due to this, individual training on sparse regions such as this may improve the overall predictions as the more consistent LSOA sizes could lead to a more representative spatial decay in feature scores. Here, it is also shown that there are not many top 25% LSOAs within this area (only 2.3%). This is because the calculation of the top 25% classification is using all the LSOAs of England and Wales. London is a stagnant region, with the lowest annual price change in 2025 (2.3%) and the highest monthly price change (2.3%) [40]. This smallest annual price change implies that the value of London is already high and therefore has a lower probability of increasing in value. In addition, the first-time buyer persona should avoid this area with its no flats or maisonettes rule removing the majority of properties that fall into a reasonable price range for the first-time buyer. If a house is purchased in this area, it will also likely be leasehold rather than freehold, potentially decreasing the expected value increase as the leasehold expiry date draws closer.

Within the region of London, the predictions displayed a much lower precision than that of other regions, such as Bristol. This is likely because the densely populated capital has a vast transport network, multiple large and prestigious universities, and a high business score. This means, that since the XGBoost model predicts the unknown region and relies more on spatial data to determine scores (as found in

Chapter 6) this area with extremely high spatial scores will look more promising, leading to more top 25% classifications. This is seen in Table 7.2, where XGBoost decreased in precision with the introduction of spatial features.

7.3 Feature Explanation



(a) SHAP beeswarm plot showcasing the top 15 features in the XGBoost regression model

(b) SHAP beeswarm plot showcasing the top 15 features in the XGBoost direct classifier model

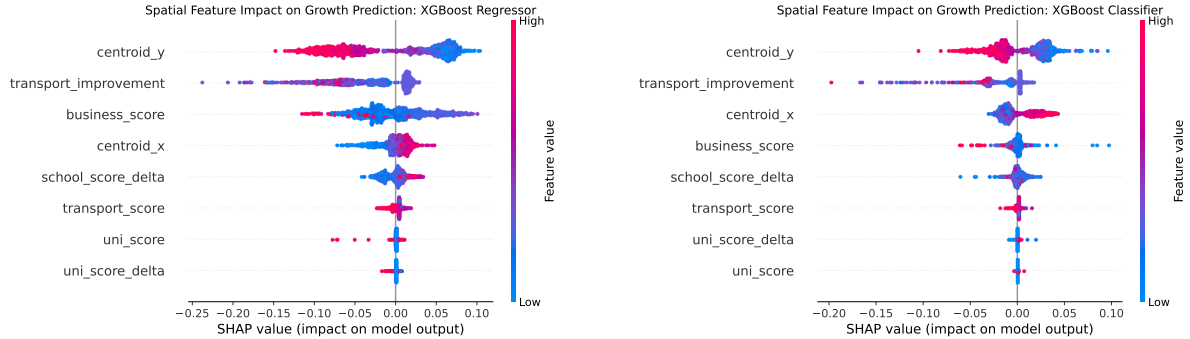
Figure 7.4: A comparison of the top features in both regression and classification predictions using an XGBoost model and the test set

Fig 7.4 compares the top 15 features from both the regression and classification XGBoost models using a beeswarm plot, which visualizes feature importance by plotting individual data points to show both their impact magnitude and their distribution density. Here, it is shown that the regression model relies on both spatial and financial data, whereas the direct classifier focuses only on financial data. This explains the change in recall and precision for the regress-then-classify model and the minimal impact on the direct classifier with the introduction of spatial features, as discussed in Chapter 6. Both models appear to also rank the financial features closest to the prediction anchor year (“Y3_Q4” features) and furthest from the prediction year (“Y1_Q1” features) highly. This is likely due to “Y3” features holding the most up to date information about the area, making them the most relevant for a three year future prediction from this anchor and “Y1” holding data from three years before the prediction anchor year, providing an example of how data in that given area has previously moved over a similar timespan.

Both models rank “inflPrice.Y3.Q4” (the $TruePrice_{LSOA,t}$ for the final quarter before the prediction anchor) within the top two features. This shows that the raw price data of previous years is one of the most important factors in predicting future area price increases. More specifically, Fig 7.4 shows that in both models raw price data possesses the highest value to predictions when below the average. This means that the model has learnt that a low value area has a greater potential to increase in value compared to an already high value area. Another top performing feature within both models is the “lsoa_price_index.Y3.Q4” (the $LPI_{LSOA,t}$ for the final quarter before the prediction anchor), which also shows its greatest value when below the average. This shows that the model has learnt that if an area belonging to a LAD is below the mean price, it is more likely to increase in value. Alternatively, it is also shown by the other “lsoa_price_index” ($LPI_{LSOA,t}$) that if it is not the final quarter before the prediction anchor, the feature is in fact more valuable if it is above the LAD mean, potentially implying that the model uses older price indices to identify established growth momentum within a district. If a LSOA is consistently over the LAD mean it signals desirability, unlike in the final quarter where it implies saturation and a lower likelihood of further inflation

A final observation from both SHAP plots in Fig 7.4 is the use of “price_delta.yoy.Y3.Q4” (the $G_{yoy,LSOA,t}$ for the final quarter before the prediction anchor), a prominent feature in both plots; it also deems lower scores to hold higher value. This suggests that if an area is slowing or decreasing in value,

relative to a year ago, it indicates a likelihood of value appreciation over the course of the next three years. This implies real estate follows a cyclic trend, and as an area drops in value, it has more space in which to increase, taking advantage of potential market overcorrections. This is an advanced concept that Pyhr et al. [33] claim was both an area of controversy between professors in 1999 and an area of high importance in predictions. The fact that the model can both identify the importance of cyclic trends and use this non-intuitive cyclic metric showcases the sophistication of the top model.



(a) SHAP beeswarm plot showcasing the spatial features in the XGBoost regression model (ranked by importance)

(b) SHAP beeswarm plot showcasing the spatial features in the XGBoost direct classifier model (ranked by importance)

Figure 7.5: A comparison of the spatial features importance in both regression and classification predictions using an XGBoost model and the test set

Fig 7.5 compares the feature importance of spatial features for both models. Here we see that the latitudinal and longitudinal coordinates, $TransportScore_{\Delta,LSOA,T}$ and $BusinessScore_{LAD,T}$ are the most valuable features, while $UniversityScore_{LAD,T}$ and its delta alternative are the least valuable features. The most valuable feature, the latitudinal coordinates, claim that southern areas are typically more likely to increase in value over areas in the north. Although this is a fact that was true throughout the years of the training data (1995 – 2012), it has recently changed, with the North East and North West having the highest annual change in 2025 (9.1% and 6.8% respectively) [40]. This is a possible reason for the low regression and classification scores in the test set (2018 – 2022) compared to the validation set (2013 – 2017), as shown later in Fig 7.7. Using the longitudinal coordinates, the model also classifies the East (where London, Oxford and Cambridge reside) as more valuable than the West (where Wales resides).

Finally, $BusinessScore_{LAD,T}$ and $TransportScore_{\Delta,LSOA,T}$ are likely two scores with a dependency between them, with the ability to reach a business district being dependent on the transport facilities of an area. $TransportScore_{\Delta,LSOA,T}$ is shown to have the most value when it has a negative SHAP value. This is because many landlords pre-emptively buy up areas in which new transport facilities are going to be built, and so by the time the facility has been built, causing the $TransportScore_{\Delta,LSOA,T}$ to increase, the value of the area has already been saturated, adding little value for the subsequent years until this score decreases and real-estate prices readjust. It was also found that an area with little business activity is more valuable, implying many want to live outside of the central business district (areas such as Surrey), or in extreme cases (the points with the most value), live in the countryside.

Comparing an ablation study of the base features’ impact on R^2 score in the test set, as shown in Fig 7.6, with the ablation in the validation set (Fig 6.1a), it can be noted that the “business score” (a top 4 feature in the validation set) has a negative impact when used in the test set. This is due to Covid-19, an event that starts in the initial years of the test set. This forced people to work from home and later on hybrid work schedules, the current workflow for many companies. Due to this change in business practice, the distance to a central business district became a less important factor in real-estate pricing. This is likely another reason why the validation set outperforms the test set to such a level, as shown in Fig 7.7.

Comparing the results of the test set and the validation set shown in Fig 7.7 it is clear that the model performs better on the validation set. This is expected, as test data is kept isolated during the training process, while parameters are tuned based on the predictions of the validation set. However, despite this expectation, there are major discrepancies, greater than just a tuning bias between the two sets. This is particularly evident in the regression model, where a more complex model is utilised. This is likely due to the data being split chronologically to ensure look-ahead bias did not occur. Although this restricts

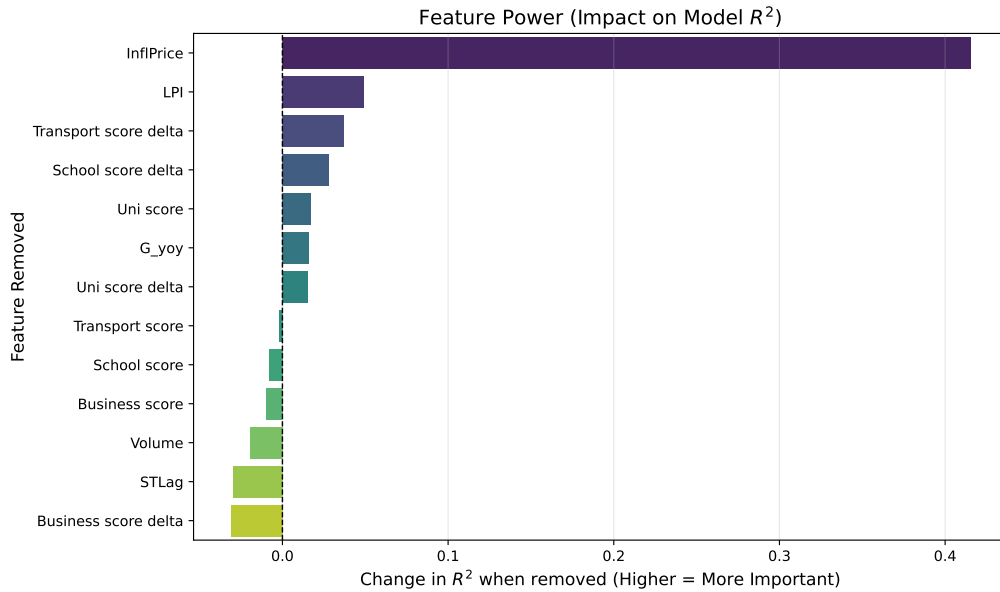


Figure 7.6: Ablation study detailing the impact of feature removal on the R^2 score of the test set

the model’s ability to cheat, it also means predictions for the years 2022 – 2025 are using trends learnt from predicting years 2001 – 2018, meaning at best this split assumes real-estate trends did not change between 2019 – 2021, years where there was a global pandemic. As evidenced by the shifting influence of features, such as latitudinal coordinates and business scores, this shift has likely resulted in performance metrics that under-represent the model’s true capabilities when applied to more recent datasets.

7.4 Adaptation for Other User Types

7.4.1 First-Time Buyer

This is the primary persona of this paper, and so the properties and build types have already been filtered for this goal. However, currently the benchmarked models have been adapted for an optimal

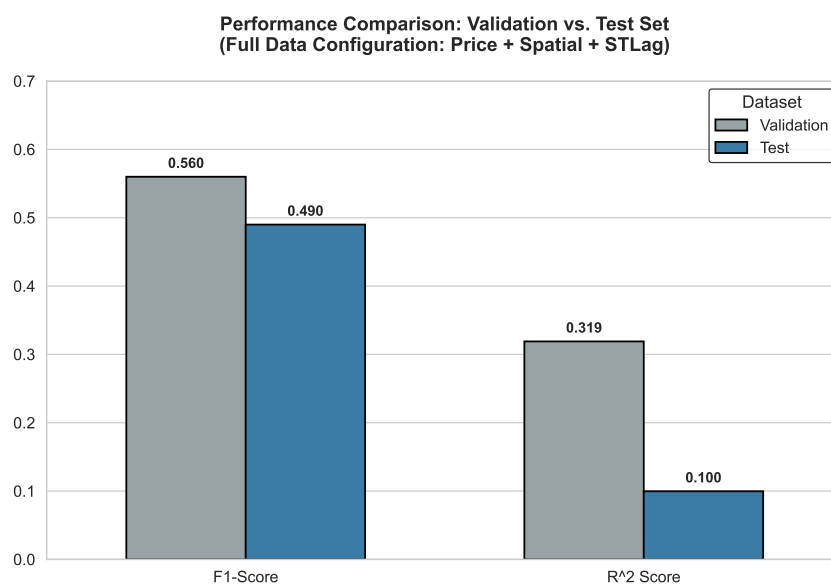


Figure 7.7: Comparison of the Regress-Then-Classify and Direct Classifier pipeline on the test set using XGBoost regressor and XGBoost direct classifier

F1-score, and therefore a balance between precision and recall. For a first-time buyer, precision is the primary goal since they are purchasing a singular property with the majority of their money invested. Therefore, in the precision-recall trade-off, they will most likely sacrifice identifying more top performing areas for the reduction in overall risk. To reflect this, a lower `scale_pos_weight` of 0.5 was chosen, as a lower `scale_pos_weight` penalises false positives more heavily, producing the confusion matrix shown in Fig 7.8a.

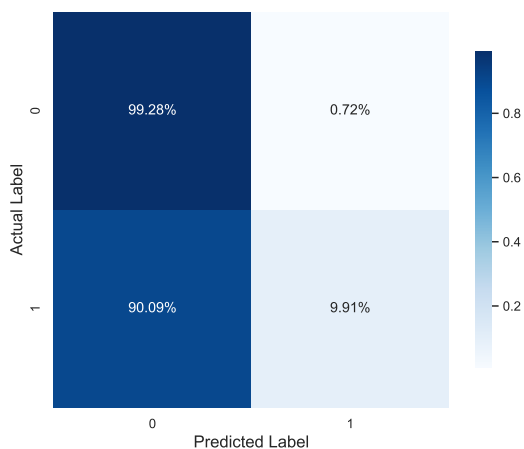
Comparing the confusion matrix in Fig 7.8a to that of the non-optimised XGBoost direct confusion matrix in Fig 6.4b, it is shown that although 85% of top 25% areas were lost, the chance that a top area was misclassified drops by 98% (5 false positives and 25 hits per 250 top LSOAs), as calculated in Appendix C. Although this is an extreme example, it shows that the first-time buyer can tune and decrease this value to match their personal acceptable level of risk.

7.4.2 Institutional Investor

For an institutional investor, someone who has a large sum of money and has the ability to invest in multiple areas, the identification of top 25% areas is likely more important with these investors willing to sacrifice precision to identify more high-growth areas. For this persona, a higher `scale_pos_weight` is more beneficial, penalising false negatives more heavily. To showcase this, a `scale_pos_weight` of 5 is shown in Fig 7.8b.

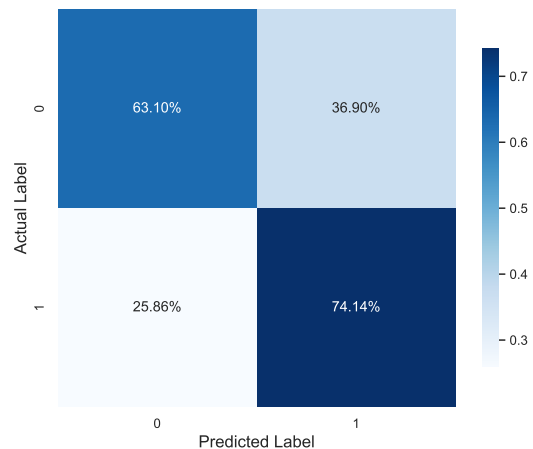
Comparing the confusion matrix in Fig 7.8b to that of the non-optimised XGBoost direct confusion matrix (Fig 6.4b), it is shown that although there is a 31% increase in false positives, 11% more top 25% LSOAs are identified (277 false positives and 185 hits per 250 top LSOAs), as calculated in Appendix C. Although a 31% increase in false positives may not be worth the 11% more LSOAs, and the currently selected (first-time buyer focused) housing dataset may not be reflective of an institutional investor's desired portfolio, it does show that the classifier possesses the ability to be tuned to a specific users desired risk levels.

Confusion Matrix for a First-Time Buyer Optimised Model



(a) Confusion Matrix showcasing an XGBoost direct classifier with a `scale_pos_weight` of 0.5 (optimised for the first-time buyer persona)

Confusion Matrix for an Institutional Investor Optimised Model



(b) Confusion Matrix showcasing an XGBoost direct classifier with a `scale_pos_weight` of 5 (optimised for the institutional investor persona)

Chapter 8

Conclusion

8.1 Summary

This paper presents a pipeline consisting of an XGBoost regression model and an XGBoost direct classification model, identified as the optimal pipeline for assisting first-time buyers (see Chapter 7). Here, the XGBoost regression model was shown to be the most reliable predictor for the percent change of LSOAs, as shown in Fig 7.1 where it displays the closest representation to the test data's mean and standard deviation. The classifier demonstrates a strong ability to effectively identify the top 25% of LSOAs with a precision 1.6 times greater than that of a random baseline and a recall that captures two-thirds of all high-growth areas. Furthermore, the system exhibits significant risk-adaptability, as discussed in Section 7.4 where precision could be raised to 0.79 at the expense of recall.

This optimal configuration was determined through the development of four distinct regression models (see Chapter 4) and the subsequent implementation of various classification pipelines built on these base models (see Chapter 5). A rigorous comparative analysis was then performed to evaluate the performance of the pipelines (see Chapter 6) and the underlying models (see Chapter 7), ultimately identifying the optimal architecture for the final system.

A comprehensive dataset was successfully compiled, spanning a 30-year period from 1995 to 2025 (see Chapter 4). This dataset integrates the financial details of each LSOA along with the impact of the local infrastructure; consisting of schools, transportation networks, business hubs, and universities. The predictive utility of this data was later validated through its application in models, which yielded high-level performance in real estate forecasts.

8.2 Future Work

8.2.1 Adaptive Bandwidths for Density-Aware Spatial Decay

As discussed in Chapter 7, the model was found to misclassify smaller LSOAs, implying the naive spatial decay approach of taking all neighbours within a set radius created a weak signal; identifying the need for a more sophisticated adaptive bandwidth. For this, two methods have been identified.

Approach A: Adaptive Radial Decay

This approach involves dynamically adjusting the search radius based on the local spatial density. By adapting the dataset to hold the population density of each LSOA, the model can scale its bandwidth: constricting the radius in dense urban areas to capture hyper-local signals and expanding it in sparse rural areas to account for larger catchments. This ensures that the decay function remains representative of the service area of local amenities regardless of geographic scale.

Approach B: Adjacency-Based Decay

Currently, I identify a target's neighbours through querying a `KDTree` with a given fixed bandwidth, returning all neighbours within that radius. This query can be easily adapted for topological decay by using functionality already present in the `KDTree` library to instead find the target's x nearest neighbours. The benefit of this approach is that it requires minimal alteration to the dataset, making use of the LSOA's

already consistent population size to model the higher friction of movement in urban environments without requiring additional density metadata.

8.2.2 Gated Ensemble Modelling

Comparing Fig 7.2 and Fig 7.3, it is evident that different regions have various levels of sparsity. In the current pipeline, one model is used to learn the behaviour of both sparse and dense areas simultaneously, leading to over-generalisation and the loss of finer, environment-specific, details. Consequently, I suggest that future work should split training into two models, training one in sparse areas, such as Norfolk, and training the other in dense areas, such as London.

Within this two-model framework, a gated ensemble can be employed to dynamically weight predictions. The gating mechanism would thereby observe a metric of local sparsity, such as the target area's decided bandwidth (if adaptive radial decay is implemented) or distance to the furthest neighbour (if adjacency-based decay is implemented) to determine the optimal influence of each model. This allows for a continuous transition between urban and rural logic, ensuring that the final output is specialised to the local geographic context.

8.2.3 Online Learning and Temporal Windows

Through the empirical analysis in Chapter 7 it was evident that real estate is a dynamic market where the influence of signals changes over time, and so legacy signals, such as those from 1995, have diminishing relevance for modern trend analysis, whereas signals from current years, such as 2025, will better demonstrate the current impact of features. To mitigate the impact of stagnant signals, I suggest the implementation of an online learning system. Here, the dataset should be updated annually and a sliding window applied to ensure only the ten most recent years are used to train the model, thereby minimising the risk of over-generalisation by filtering out the noise of legacy signals.

Bibliography

- [1] Jon Louis Bentley. Multidimensional binary search trees used for associative searching. *Commun. ACM*, 18(9):509–517, sep 1975. doi:[10.1145/361002.361007](https://doi.org/10.1145/361002.361007).
- [2] Karl E. Case and Robert J. Shiller. Forecasting prices and excess returns in the housing market. Technical Report Working Paper 3368, National Bureau of Economic Research, May 1990. NBER Working Paper. URL: <https://www.nber.org/papers/w3368>, doi:[10.3386/w3368](https://doi.org/10.3386/w3368).
- [3] Vitor Cerqueira, Luis Torgo, Fabio Pinto, and Carlos Guedes Soares. Predicting wave heights: A comparison of direct and indirect forecasting approaches, 2024. URL: <https://arxiv.org/abs/2206.09821>, arXiv:2206.09821.
- [4] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, pages 785–794, New York, NY, USA, 2016. ACM. doi:[10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785).
- [5] Curvo. Ftse 100 market index backtest (currency = eur). <https://curvo.eu/backtest/en/market-index/ftse-100?currency=eur>, 2026. Accessed: 2026-02-10; historical performance/backtest of the FTSE 100 index in euros.
- [6] Department for Education. Get information about schools. GOV.UK Service, 2024. Accessed: 2024-04-17. URL: <https://get-information-schools.service.gov.uk/Downloads>.
- [7] Department for Transport. National public transport access nodes (naptan). GOV.UK NaPTAN Portal, 2024. Accessed: 2024-04-17. URL: <https://beta-naptan.dft.gov.uk/download/national>.
- [8] Andrew Ebekozen, Abdul-Rashid Abdul-Aziz, and Mastura Jaafar. Housing finance inaccessibility for low-income earners in malaysia: Factors and solutions. *Habitat International*, 87:27–35, 2019. URL: <https://www.sciencedirect.com/science/article/pii/S0197397518308142>, doi:[10.1016/j.habitatint.2019.03.009](https://doi.org/10.1016/j.habitatint.2019.03.009).
- [9] Stewart A. Fotheringham, Chris Brunsdon, and Martin Charlton. *Geographically Weighted Regression: The Analysis of Spatially Varying Relationships*. John Wiley & Sons, Chichester, UK, 2002.
- [10] GOV. Live tables on dwelling stock: Table 125 - dwelling stock estimates by local authority district. GOV.UK Statistical Data Set, 2024. Accessed: 2024-04-17. URL: <https://www.gov.uk/government/statistical-data-sets/live-tables-on-dwelling-stock-including-vacants>.
- [11] Jennie Gray, Lisa Buckner, and Alexis Comber. Predicting gentrification in england: A data primitive approach. *Urban Science*, 7(2):64, 2023. URL: <https://www.mdpi.com/2413-8851/7/2/64>, doi:[10.3390/urbansci7020064](https://doi.org/10.3390/urbansci7020064).
- [12] Jake Grigsby, Zhe Wang, Nam Nguyen, and Yanjun Qi. Long-range transformers for dynamic spatiotemporal forecasting. In *The Twelfth International Conference on Learning Representations*, 2024. URL: <https://openreview.net/forum?id=Yp3hY9uI7S>.
- [13] Higher Education Statistics Agency. He student data: Table 1 - he students by he provider, level of study, mode of study and domicile. HESA Data and Analysis, 2024. Accessed: 2024-04-17. URL: <https://www.hesa.ac.uk/data-and-analysis/students/table-1>.
- [14] HM Land Registry. Price paid data. Open Government Licence, 2024. Accessed: 2024-04-17. URL: <https://www.gov.uk/government/statistical-data-sets/price-paid-data-downloads>.

- [15] Bo Huang, Bo Wu, and Michael Barry. Geographically and temporally weighted regression for modeling spatio-temporal variation in house prices. *International Journal of Geographical Information Science*, 24(3):383–401, 2010. doi:10.1080/13658810802672469.
- [16] Muharem Karamujic. *The Global Importance of Housing and the Emergence of the Housing Affordability Problem*, pages 7–26. Palgrave Macmillan UK, London, 2015. URL: https://link.springer.com/chapter/10.1057/9781137517937_2, doi:10.1057/9781137517937_2.
- [17] Michael H. Kutner, Christopher J. Nachtsheim, John Neter, and William Li. *Applied Linear Statistical Models*. McGraw-Hill/Irwin, New York, 5 edition, 2005.
- [18] H. Li et al. Analyzing housing prices in shanghai with open data: Amenity, accessibility and urban structure. *Cities*, 91:165–179, 2019. URL: <https://www.sciencedirect.com/science/article/pii/S0264275118305936>, doi:10.1016/j.cities.2018.11.016.
- [19] Macrotrends. S&p 500 historical annual returns (1927–2026). <https://www.macrotrends.net/2526/sp-500-historical-annual-returns>, 2026. Accessed: 2026-02-10; historical annual return data for the S&P 500 index.
- [20] Nominatim Contributors. *Nominatim: Open-source Search and Geocoding Engine*, 2024. Version 4.4.0. URL: <https://nominatim.org/>.
- [21] Office for National Statistics. Business demography, 2009. UK Government Web Archive, 2010. Accessed: 2024-04-17. URL: <https://webarchive.nationalarchives.gov.uk/ukgwa/20151013215343/http://www.ons.gov.uk/ons/rel/bus-register/business-demography/2009/index.html>.
- [22] Office for National Statistics. Lower layer super output areas (december 2021) boundaries ew bgc v5. ONS Geography Portal, 2021. Accessed: 2024-04-17. URL: <https://geoportal.statistics.gov.uk/datasets/ons::lower-layer-super-output-areas-december-2021-boundaries-ew-bgc-v5-2/about>.
- [23] Office for National Statistics. Lsoa (2011) to lsoa (2021) to local authority district (2022) lookup in england and wales. ONS Geography Portal, 2022. Accessed: 2024-04-17. URL: <https://geoportal.statistics.gov.uk/datasets/3be72478d8454b59bb86ba97b4ee325b/about>.
- [24] Office for National Statistics. Business demography, uk: reference table. ONS Statistical Bulletin, 2024. Accessed: 2024-04-17. URL: <https://www.ons.gov.uk/businessindustryandtrade/business/activitysizeandlocation/datasets/businessdemographyreferencetable/current>.
- [25] Office for National Statistics. Consumer price inflation time series (mm23). ONS Statistical Bulletin, 2024. Accessed: 2024-04-17. URL: <https://www.ons.gov.uk/economy/inflationandpriceindices/datasets/consumerpriceindices>.
- [26] Office for National Statistics. Uk business counts / business demographics (1995-2007). Nomis Official Labour Market Statistics, 2024. Data retrieved via Nomis Query Constructor. Accessed: 2024-04-17. URL: <https://www.nomisweb.co.uk/>.
- [27] Office for National Statistics. Statistical geographies. <https://www.ons.gov.uk/methodology/geography/ukgeographies/statisticalgeographies>, 2026. Accessed: 2026-02-24; overview of the statistical geographies used across the UK for official statistics.
- [28] OpenDataSoft. Regions – united kingdom (georef-united-kingdom-region) [dataset]. <https://public.opendatasoft.com/explore/dataset/georef-united-kingdom-region>, 2026. Accessed: 2026-02-24; part of the Geographical repository maintained by OpenDataSoft; contains data for regions in the United Kingdom.
- [29] Byeonghwa Park and Jae Kwon Bae. Using machine learning algorithms for housing price prediction: The case of Fairfax County, Virginia housing data. *Expert Systems with Applications*, 42(6):2928–2934, 2015. doi:10.1016/j.eswa.2014.11.040.
- [30] Chris Pettit, Y Shi, H Han, M Rittenbruch, M Foth, S Lieske, R van den Nouwelant, P Mitchell, S Leao, B Christensen, and M Jamal. A new toolkit for land value analysis and scenario planning. *Environment and Planning B: Urban Analytics and City Science*, 47(8):1490–1507, 2020. doi:10.1177/2399808320924678.

- [31] Planning Data. Local authority district [dataset]. <https://www.planning.data.gov.uk/dataset/local-authority-district>, 2026. Accessed: 2026-02-24; administrative boundaries of local authority districts in England provided by the Office for National Statistics under the Open Government Licence v3.0.
- [32] Henry O. Pollakowski and Susan M. Wachter. The effects of land-use constraints on housing prices. *Land Economics*, 66(3):315–324, 1990. URL: <https://www.jstor.org/stable/3146732>, doi:10.2307/3146732.
- [33] Stephen A. Pyhrr, Stephen E. Roulac, and Waldo L. Born. Real estate cycle and market analysis. *Journal of Real Estate Research*, 18(1):5–32, 1999.
- [34] Darren P. Smith. ‘studentification’: The gentrification factory? In Rowland Atkinson and Gary Bridge, editors, *Gentrification in a Global Context: The New Urban Colonialism*, chapter 5, pages 72–89. Routledge, London, 2005.
- [35] Ali Soltani, Mohammad Heydari, Fatemeh Aghaei, and Christopher James Pettit. Housing price prediction incorporating spatio-temporal dependency into machine learning algorithms. *Cities*, 131:103941, 2022. doi:10.1016/j.cities.2022.103941.
- [36] Luka Straš, Matjaž Gams, and Sebastjan Kocjan. Radonfan: Intelligent real-time radon mitigation through iot, rule-based logic, and ai forecasting. *AI*, 7(2):67, 2026. URL: <https://www.mdpi.com/2673-2688/7/2/67>, doi:10.3390/ai7020067.
- [37] Ilya Sutskever, Oriol Vinyals, and Quoc V. Le. Sequence to sequence learning with neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27. Curran Associates, Inc., 2014. URL: <https://arxiv.org/abs/1409.3215>, arXiv:1409.3215.
- [38] Yiqiu Tang, Yanjun Chen, and Shenghan Zhao. Regional housing supply and demand imbalance: Qualitative analysis in u.s. based on big data. In *Proceedings of the BDEIM 2024 (Big Data Economy and Information Management)*. Association for Computing Machinery, 2025. doi:10.1145/3724154.3724255.
- [39] Quang Truong, Minh Nguyen, Hy Dang, and Bo Mei. Housing price prediction via improved machine learning techniques. *Procedia Computer Science*, 174:433–442, 2020. doi:10.1016/j.procs.2020.06.111.
- [40] UK Government Office for National Statistics and HM Land Registry. Uk house price index summary: January 2025. Technical report, UK Government, March 2025. Accredited official statistics report including headline UK house price index data and regional breakdowns. URL: <https://www.gov.uk/government/statistics/uk-house-price-index-for-january-2025/uk-house-price-index-summary-january-2025>.
- [41] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pages 5998–6008, 2017. URL: <https://arxiv.org/abs/1706.03762>.
- [42] Pei-Ying Wang, Chiao-Ting Chen, Jain-Wun Su, Ting-Yun Wang, and Szu-Hao Huang. Deep learning model for house price prediction using heterogeneous data analysis along with joint self-attention mechanism. *IEEE Access*, 9:55269–55279, 2021. doi:10.1109/ACCESS.2021.3071305.
- [43] Feng Yuan et al. Amenity effects of urban facilities on housing prices in china. *Land Use Policy*, 95:104639, 2020. URL: <https://www.sciencedirect.com/science/article/pii/S0264275118304700>, doi:10.1016/j.landusepol.2020.104639.
- [44] Nor Hamizah Zulkifley, Shuzlina Abdul Rahman, Nor Hasbiah Ubaidullah, and Ismail Ibrahim. House price prediction using a machine learning model: A survey of literature. *International Journal of Modern Education and Computer Science (IJMECS)*, 12(6):46–54, 2020. URL: <https://www.mecspress.org/ijmeecs/ijmeecs-v12-n6/v12n6-4.html>, doi:10.5815/ijmeecs.2020.06.04.

Appendix A

Appendix A: AI Prompts/Tools

Verifying Understanding of Papers

With difficult aspects of papers, such as the encodings of Spacetimeformer [12] and XGBoost's [4] non-intuitive jumps between formulas, Gemini was used to confirm understanding before re-reading aspects of the paper it highlighted and asking again:

```
>>> Is the logic for my explanation of Spacetimeformer correct?
Here's the paper: [link] and the explanation: [text]
```

Converting the Python Version of GTWR into R

After coding and running my first iteration of GTWR in python I realised that GTWR was been depreciated, due to being less familiar with R and as I had already written the structure in python I asked the following:

```
>>> I am getting ModuleNotFoundError: No module names 'stwr'
```

It's depreciated in python, you can do it in R

```
>>> Can you convert this python code into R: [code]
```

after this Gemini was used to help me understand aspects of R such as parallelising GTWR

Finding Datasets for Spatial Features

It was very hard to find 30 years worth of data for certain features, due to this Gemini was used to help find possible datasets:

```
>>> Is there a database listing all the Central buisness districts in England ↩
and Wales between 1995 and 2025, at either a LSOA or LAD level
```

There aren't set CBDs in the UK but you can use this Business Structure Database

```
>>> This is restricted access due to some rows being business identifiable, is↩
there a simpler version with birth and death rates only
```

Here's one for years between 2010 and 2024

After finding this set I looked in a lot of archives and sent suggestions to Gemini until I got all the years I needed.

Checking Grammar and Logic

To make sure my grammar and logic of certain paragraphs where correct I used Gemini to check it:

```
>>> Do not show a rewrite. Look at my logic and grammar in this paragraph and ↩
suggest changes: [text]
```

Creating Graphs

To quickly generate graphs I got Gemini to construct the python code for me then altered it to get what I want:

```
>>> Can you make a bar chart comparing the R^2 of MLR, XGBoost and ↵  
      Spacetimeformer, here is the data: [data]
```

Appendix B

Further Technical Background

B.1 SeqToSeq

Deep neural networks, although powerful, can only be applied to problems whose inputs and targets can be encoded with vectors of fixed dimensionality. However, for tasks such as speech recognition the sequence length is not known prior, therefore there is a need for a domain-independent method that learns to map sequences to sequences. Seq2Seq is one such method; making use of Long Short-Term Memory (LSTM) architecture, it aims to estimate the conditional probability $p(y_1, \dots, y_{T'} | x_1, \dots, x_T)$ where $(x_1, \dots, x_T)$ is an input sequence and $(y_1, \dots, y_{T'})$ is its corresponding output sequence whose length T' may differ from T [37]. It does this by obtaining the fixed-dimensional representation v of the input sequence $(x_1, \dots, x_T)$ given by the last hidden state of the LSTM and then computing the probability of $y_1, \dots, y_{T'}$ with a standard LSTM-LM formulation whose initial hidden state is set to the representation v of $x_1, \dots, x_T$ [37]:

$$p(y_1, \dots, y_{T'} | x_1, \dots, x_T) = \prod_{t=1}^{T'} p(y_t | v, y_1, \dots, y_{t-1}) \quad (\text{B.1})$$

In the above equation, each $p(y_t | v, y_1, \dots, y_{t-1})$ distribution is represented with a softmax over all the words in the vocabulary, and each sentence was required to end with a special end-of-sentence symbol “<EOS>” [37]. Finally, the specific Seq2Seq architecture makes use of two separate LSTMs: one for the input sequence and the other for the output sequence. It also uses a deep LSTM of four layers as it found this outperformed shallow LSTMs [37].

B.2 KDTree

B.2.1 Structure

The following mathematical framework for the KDTree in this section is derived from the foundational work of Bentley [1]. KDTree is a data structure for associative searching. If a file is stored as a tree, each record is a node in the tree that contains two pointers pointing to either null or to another node. A discriminator is associated with each node in this binary k -depth tree, this value is an integer between 0 and $k - 1$ that showcases the nodes depth. On this basis k keys of node P will be defined as $K_0(P), \dots, K_{k-1}(P)$, pointers will be $LOSON(P)$ and $HISON(P)$, and the discriminator will be $DISC(P)$. The defining order of this tree is as follows: “for any node P in a $k - d$ tree, let j be $DISC(P)$, then for any node Q in $LOSON(P)$, it is true that $K_j(Q) < K_j(P)$; likewise, for any node R in $HISON(P)$, it is true that $K_j(R) > K_j(P)$ ” [1] (in simpler terms $LOSON$ refers to children left of node P and $HISON$ refers to children right of node P).

$$NEXTDISC(i) = (i + 1) \mod k \quad (\text{B.2})$$

With this definition, I can state that, in general, $NEXTDISC$ (Equation B.2) defines the direct children of a node, i.e. $NEXTDISC(DISC(P)) = DISC(LOSON(P)) = DISC(HISON(P))$.

B.2.2 Distance Calculations

$$S_j(P) = K_j(P)K_{j+1}(P) \dots K_{k-1}(P)K_0(P) \dots K_{j-1}(P) \quad (\text{B.3})$$

$SUCCESSOR(P, Q)$ is the next node from P to visit to reach Q . For this I must use the superkey defined in Equation B.3, if $S_j(Q) < S_j(P)$, then $SUCCESSOR$ returns $LOSON$ and if $S_j(Q) > S_j(P)$, it returns $HISON$. If $S_j(Q) = S_j(P)$ then all k keys are equal and $SUCCESSOR$ returns a special value to indicate that. With a definition of $SUCCESSOR$ I can define node Q to be j -greater than node P iff $SUCCESSOR(P, Q)$ is $HISON$, and j -less iff $SUCCESSOR(P, Q)$ is $LOSON$ [1]. This can be used to find j -maximum and j -minimum elements of a collection of nodes. The j -distance between nodes P and Q is $|K_j(P) - K_j(Q)|$.

B.2.3 Region Queries

```

Boolean Procedure      : IN_REGION
                        (node P)
for I  $\leftarrow$  0 step 1 until k - 1 do
    | if  $K_I(P) < RECDEF(2 \cdot I)$  then
    | | return false
    | end
    | if  $K_I(P) > RECDEF(2 \cdot I + 1)$  then
    | | return false
    | end
end
return true

```

Algorithm B.0: Pseudo-Algol IN_REGION: returns true iff P is in the hyper-rectangle defined by $RE CDEF$ [1]

```

Boolean Procedure: BOUNDS_INTERSECT_REGION
    (array  $B$ )
for  $I = 0$  step 2 until  $2 \cdot (k - 1)$  do
    | if  $B(I) > RECDEF(I + 1)$  then
    | | return false
    | end
    | if  $B(I + 1) > RECDEF(I)$  then
    | | return false
    | end
end
return true

```

Algorithm B.1: Pseudo-Algorithm
BOUNDS_INTERSECT_REGION: returns true
iff the hyper-rectangle defined by bounds array B intersects the hyper-rectangle defined by $RECDEF$ [1]

Region queries are queries that retrieve records that intersect a specified region. Specifically, *REGIONSEARCH* is the form of region query used in spatial score compilation in Chapter 4. This is a recursive algorithm that takes a root node (P) and radius (bounds array B) and then uses the bounds stored at each node of the tree to determine whether it is possible that any descendants of the node may lie in the region being searched, thereby visiting as few nodes as possible. It then reports all nodes in the subtree whose root is P which are in the region by invoking the *FOUND* procedure [1].

1. [Is P in the region?] If $IN_REGION(P)$ then call $FOUND(P)$.
2. [Improve the bounds for the subtrees] Allocate B_L , B_H as bounds arrays, and copy the array B into both. Set $J \leftarrow DISC(P)$ (J is the dimension of the bounds to be changed). Set $B_L(2J+1) \leftarrow K_J(P)$, $B_H(2J) \leftarrow K_J(P)$ (this step notes that $K_J(P)$ is a J -upper bound of the nodes in the $LOSON$ subtree and a J -lower bound of the nodes in the $HISON$ subtree).
3. [Search $LOSON$ subtree] If $LOSON(P) \neq \Lambda$ and $BOUNDS_INTERSECT_REGION(B_L)$ then $REGIONSEARCH(LOSON(P), B_L)$
4. [Search $HISON$ subtree] If $HISON(P) \neq \Lambda$ and $BOUNDS_INTERSECT_REGION(B_H)$ then $REGIONSEARCH(HISON(P), B_H)$

B.2.4 Nearest Neighbour Queries

This is the form of query suggested in future work (Section 8.2). Here, given a distance function D , a collection of points B , and a point P , P 's nearest neighbour Q in collection B is defined such that:

$$(\forall R \leq B) \{ (R \neq Q) \Rightarrow [D(R, P) \geq D(Q, P)] \} \quad (\text{B.4})$$

Appendix C

Features And Extra Results

C.1 Features

Table C.1: Description of features

Category	Feature	Description	Loc.	Freq.
Financial	<i>TruePrice</i>	Inflation-adjusted median price	LSOA	Qrt.
	<i>Volume</i>	Total number of property sales	LSOA	Qrt.
	<i>G_{yoy}</i>	Year-on-year % change in <i>TruePrice</i>	LSOA	Qrt.
	<i>LPI</i>	Local Authority District price index	LSOA	Qrt.
Spatial	<i>SchoolScore</i>	LSOA desirability based on school data	LSOA	Yrly.
	<i>SchoolScore_Δ</i>	3-year trend in school desirability	LSOA	Yrly.
	<i>UniScore</i>	Level of studentification within the LAD	LAD	Yrly.
	<i>UniScore_Δ</i>	3-year trend in studentification	LAD	Yrly.
	<i>BusinessScore</i>	Employment and job density within the LAD	LAD	Yrly.
	<i>BusinessScore_Δ</i>	3-year trend in job density	LAD	Yrly.
	<i>TransportScore</i>	General connectivity (NaPTAN metrics)	LSOA	Yrly.
Spatial-Temp.	<i>TransportScore_Δ</i>	3-year trend in connectivity improvements	LSOA	Yrly.
	<i>STLag</i>	Appreciation trends in neighbouring LSOAs	LSOA	Qrt.

C.2 Results

C.2.1 XGBoost Regress-Then-Classify vs. Direct Classification Calculations

Given 1000 LSOAs there will be 250 LSOAs that are in the top 25% and 750 LSOAs that are not.

$$\begin{aligned}
 RTC_{pred\ 1} &= 250 \cdot 0.5754 + 750 \cdot 0.2259 \\
 &= 143.85 + 169.425 \\
 &\approx 144 + 169 \\
 &\approx 313
 \end{aligned} \tag{C.1}$$

Using the percentages from the confusion matrices in Fig 6.4, within the regress-then-classify predictions there are approximately 144 correctly identified LSOAs and 169 incorrectly identified LSOAs.

$$\begin{aligned}
 DC_{pred\ 1} &= 250 \cdot 0.6627 + 750 \cdot 0.2825 \\
 &= 165.675 + 211.875 \\
 &\approx 166 + 212 \\
 &\approx 378
 \end{aligned} \tag{C.2}$$

Using the percentages from the confusion matrices in Fig 6.4, within the direct classifier predictions there are approximately 166 correctly identified LSOAs and 212 incorrectly identified LSOAs.

Table C.2: Comparison of Regress-Then-Classify and Direct Classifier pipelines with 3 year predictions on the test set

Model	Features	Pipeline	Prec.	Recall	F1-Score
MLR / Logistic Regression	Price	RTC	0.39	0.54	<u>0.45</u>
		Direct	0.63	0.25	0.36
	+ Spatial	RTC	0.34	<u>0.65</u>	<u>0.45</u>
		Direct	<u>0.65</u>	0.22	0.32
	+ Spatial + STLag	RTC	0.35	0.63	<u>0.45</u>
		Direct	0.61	0.28	0.38
XGBoost	Price	RTC	<u>0.42</u>	0.53	0.47
		Direct	0.39	<u>0.67</u>	<u>0.49</u>
	+ Spatial	RTC	0.39	0.60	0.47
		Direct	0.39	0.66	<u>0.49</u>
	+ Spatial + STLag	RTC	0.41	0.58	0.48
		Direct	0.39	0.66	<u>0.49</u>

Note: All metrics refer specifically to Class 1 (Top 25% investment targets). RTC denotes the Regress-Then-Classify staged pipeline.

$$\begin{aligned}
\% \uparrow Truths_{RTC,DC} &= \frac{DC}{RTC} \\
&= \frac{166}{144} \\
&= 1.1527 = 15\%
\end{aligned} \tag{C.3}$$

$$\begin{aligned}
\% \uparrow False_{RTC,DC} &= \frac{DC}{RTC} \\
&= \frac{212}{169} \\
&= 1.2544 \approx 25\%
\end{aligned} \tag{C.4}$$

$$\begin{aligned}
\% \uparrow Total_{RTC,DC} &= \frac{DC}{RTC} \\
&= \frac{378}{313} \\
&= 1.2077 \approx 21\%
\end{aligned} \tag{C.5}$$

$$\begin{aligned}
FTB_{pred\ 1} &= 250 \cdot 0.0991 + 750 \cdot 0.0072 \\
&= 24.775 + 5.4 \\
&\approx 25 + 5 \\
&\approx 30
\end{aligned} \tag{C.6}$$

Using the percentages from the confusion matrices in Fig 7.8a, within the first-time buyer optimised predictions there are approximately 25 correctly identified LSOAs and 5 incorrectly identified LSOAs.

$$\begin{aligned}
\% \uparrow Truths_{DC,FTB} &= \frac{FTB}{DC} \\
&= \frac{25}{166} \\
&= 0.1506 \approx 85\%_{loss}
\end{aligned} \tag{C.7}$$

$$\begin{aligned}
\% \uparrow False_{DC,FTB} &= \frac{FTB}{DC} \\
&= \frac{5}{212} \\
&= 0.02358 \approx 98\%_{loss}
\end{aligned} \tag{C.8}$$

$$\begin{aligned}
II_{pred\ 1} &= 250 \cdot 0.7414 + 750 \cdot 0.3690 \\
&= 185.35 + 276.75 \\
&\approx 185 + 277 \\
&\approx 462
\end{aligned} \tag{C.9}$$

Using the percentages from the confusion matrices in Fig 7.8b, within the institutional investor optimised predictions there are approximately 185 correctly identified LSOAs and 277 incorrectly identified LSOAs.

Table C.3: Comparison of Regress-Then-Classify and Direct Classifier pipelines with 3 year predictions on the unknown regions set (London and the North West)

Model	Features	Pipeline	Prec.	Recall	F1-Score
MLR / Logistic Regression	Price	RTC	0.39	0.62	<u>0.48</u>
		Direct	<u>0.68</u>	0.30	0.41
	+ Spatial	RTC	0.36	<u>0.66</u>	0.47
		Direct	0.66	0.33	0.44
	+ Spatial + STLag	RTC	0.37	0.65	0.47
		Direct	0.64	0.34	0.44
XGBoost	Price	RTC	<u>0.33</u>	0.78	<u>0.46</u>
		Direct	0.31	0.77	0.45
	+ Spatial	RTC	0.28	0.80	0.42
		Direct	0.28	<u>0.82</u>	0.42
	+ Spatial + STLag	RTC	0.30	0.77	0.44
		Direct	0.30	0.80	0.44

Note: All metrics refer specifically to Class 1 (Top 25% investment targets). RTC denotes the Regress-Then-Classify staged pipeline.

$$\begin{aligned}
\% \uparrow Truths_{DC,II} &= \frac{II}{\frac{DC}{185}} \\
&= \frac{166}{185} \\
&= 1.114 \approx 11\%
\end{aligned} \tag{C.10}$$

$$\begin{aligned}
\% \uparrow False_{DC,II} &= \frac{II}{\frac{DC}{277}} \\
&= \frac{212}{277} \\
&= 1.3066 \approx 31\%
\end{aligned} \tag{C.11}$$

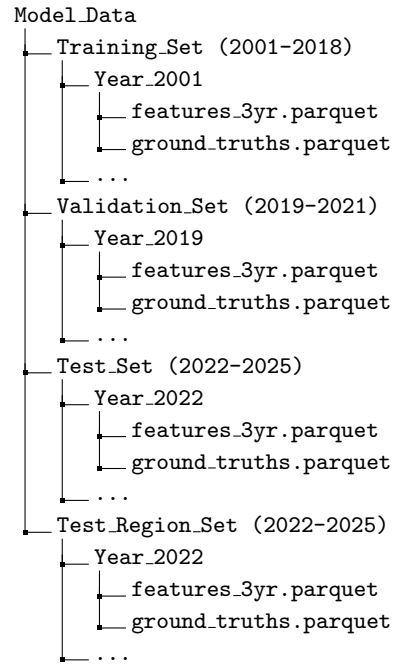


Figure C.1: Hierarchical directory structure of the processed dataset, partitioned by year to ensure temporal isolation.

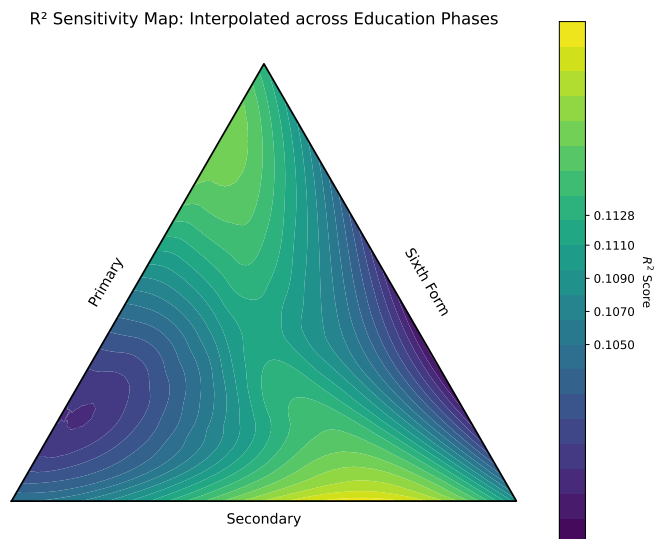


Figure C.2: Explanatory analysis of education phases' spatial impact